

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.5	4.6	1	0	0	81.9
U2 - Beta(0.5, 0.5)	13.2	4	0.5	0.1	0	82.2
U2 - Beta(0.7, 0.3)	13.1	4.1	0.4	0.1	0	82.3
U2 - Beta(1, 1)	12.9	4	0.6	0.1	0	82.4
U3 - Beta(0.3, 0.7)	13.3	3.9	0.6	0	0	82.2
U3 - Beta(0.5, 0.5)	14.4	3.4	0.4	0	0	81.8
U3 - Beta(0.7, 0.3)	14.1	3.7	0.5	0.1	0	81.6
U3 - Beta(1, 1)	13.8	3.9	0.4	0	0	81.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.4	3	1.3	0.3	0.1	-
U2 - Beta(0.5, 0.5)	11.5	2.9	1.3	0.3	0.1	-
U2 - Beta(0.7, 0.3)	11.5	2.9	1.2	0.3	0.1	-
U2 - Beta(1, 1)	11.5	3	1.2	0.3	0.1	-
U3 - Beta(0.3, 0.7)	11.5	3	1.3	0.3	0.1	-
U3 - Beta(0.5, 0.5)	11.5	2.9	1.3	0.4	0.1	-
U3 - Beta(0.7, 0.3)	11.5	3	1.2	0.3	0.1	-
U3 - Beta(1, 1)	11.5	2.9	1.2	0.3	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	1.6	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	3.4	1.6	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	3.4	1.6	0.4	0.2	0	-
U2 - Beta(1, 1)	3.4	1.6	0.4	0.2	0	-
U3 - Beta(0.3, 0.7)	3.4	1.6	0.5	0.2	0.1	-
U3 - Beta(0.5, 0.5)	3.4	1.6	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	3.4	1.6	0.4	0.2	0	-
U3 - Beta(1, 1)	3.4	1.6	0.4	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						573.7
U2 - Beta(0.5, 0.5)						573.9
U2 - Beta(0.7, 0.3)						573.6
U2 - Beta(1, 1)						574.2
U3 - Beta(0.3, 0.7)						575.1
U3 - Beta(0.5, 0.5)						573.1
U3 - Beta(0.7, 0.3)						573.2
U3 - Beta(1, 1)						574.3
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	37.5	19.7	30.3	7.2	1.1	4.2
U2 - Beta(0.5, 0.5)	37.5	20.3	30.3	6.7	1.1	4.1
U2 - Beta(0.7, 0.3)	38.1	20.3	29.5	7	1	4.1
U2 - Beta(1, 1)	37.7	19.9	29.5	7.1	1.6	4.2
U3 - Beta(0.3, 0.7)	39	19.9	29.8	6.3	0.9	4.1
U3 - Beta(0.5, 0.5)	40.5	20.5	28.2	6	0.7	4.1
U3 - Beta(0.7, 0.3)	40.9	21.7	25.9	6.6	0.6	4.3
U3 - Beta(1, 1)	41.2	20.7	27.1	6.1	0.7	4.2
Mean administrations						
U2 - Beta(0.3, 0.7)	12.4	5.6	7	3.1	1.2	-
U2 - Beta(0.5, 0.5)	12.5	5.6	6.9	3.1	1.1	-
U2 - Beta(0.7, 0.3)	12.5	5.7	6.8	3.2	1.1	-
U2 - Beta(1, 1)	12.5	5.6	6.8	3.2	1.2	-
U3 - Beta(0.3, 0.7)	12.4	5.6	7	3.1	1.2	-
U3 - Beta(0.5, 0.5)	12.5	5.6	6.9	3.1	1.2	-
U3 - Beta(0.7, 0.3)	12.5	5.7	6.8	3.2	1.1	-
U3 - Beta(1, 1)	12.5	5.6	6.8	3.2	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.6	2.6	1.9	0.9	0.3	-
U2 - Beta(0.5, 0.5)	3.7	2.6	1.9	0.9	0.3	-
U2 - Beta(0.7, 0.3)	3.7	2.6	1.9	0.9	0.3	-
U2 - Beta(1, 1)	3.7	2.6	1.9	0.9	0.3	-
U3 - Beta(0.3, 0.7)	3.7	2.6	1.9	0.9	0.3	-
U3 - Beta(0.5, 0.5)	3.7	2.6	1.9	0.9	0.3	-
U3 - Beta(0.7, 0.3)	3.7	2.6	1.9	0.9	0.3	-
U3 - Beta(1, 1)	3.7	2.6	1.9	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1095.3
U2 - Beta(0.5, 0.5)						1094.3
U2 - Beta(0.7, 0.3)						1095.1
U2 - Beta(1, 1)						1094.5
U3 - Beta(0.3, 0.7)						1095.5
U3 - Beta(0.5, 0.5)						1095.5
U3 - Beta(0.7, 0.3)						1094.4
U3 - Beta(1, 1)						1095.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.7	39.9	19.4	6.7	0.7	18.6
U2 - Beta(0.5, 0.5)	15.2	40.4	19.1	6	0.6	18.7
U2 - Beta(0.7, 0.3)	16	40.3	19.2	5.6	0.7	18.2
U2 - Beta(1, 1)	15.9	41.1	18.8	5.2	0.6	18.4
U3 - Beta(0.3, 0.7)	16.5	42.6	16.5	5.2	0.5	18.7
U3 - Beta(0.5, 0.5)	16.4	43.2	16.9	4.4	0.5	18.6
U3 - Beta(0.7, 0.3)	16.7	43.8	15.7	4.8	0.4	18.6
U3 - Beta(1, 1)	17.5	43.7	15.1	4.8	0.4	18.5
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	7.8	5.8	2.9	1.1	-
U2 - Beta(0.5, 0.5)	9.4	7.8	5.7	3	1.1	-
U2 - Beta(0.7, 0.3)	9.5	7.9	5.7	2.9	1.1	-
U2 - Beta(1, 1)	9.4	7.9	5.7	2.9	1.2	-
U3 - Beta(0.3, 0.7)	9.5	7.7	5.8	3	1.1	-
U3 - Beta(0.5, 0.5)	9.5	7.9	5.7	2.9	1.1	-
U3 - Beta(0.7, 0.3)	9.4	7.9	5.7	2.9	1.1	-
U3 - Beta(1, 1)	9.5	7.9	5.6	3	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	3.4	1.6	0.9	0.3	-
U2 - Beta(0.5, 0.5)	2.7	3.4	1.6	0.9	0.3	-
U2 - Beta(0.7, 0.3)	2.7	3.4	1.6	0.9	0.3	-
U2 - Beta(1, 1)	2.7	3.4	1.6	0.9	0.4	-
U3 - Beta(0.3, 0.7)	2.7	3.3	1.6	0.9	0.3	-
U3 - Beta(0.5, 0.5)	2.7	3.4	1.6	0.9	0.3	-
U3 - Beta(0.7, 0.3)	2.7	3.4	1.6	0.9	0.4	-
U3 - Beta(1, 1)	2.7	3.4	1.6	1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						997.9
U2 - Beta(0.5, 0.5)						997.5
U2 - Beta(0.7, 0.3)						998.9
U2 - Beta(1, 1)						998
U3 - Beta(0.3, 0.7)						996.9
U3 - Beta(0.5, 0.5)						997
U3 - Beta(0.7, 0.3)						997.1
U3 - Beta(1, 1)						997.5
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.6	9	25	42.6	11.7	4.1
U2 - Beta(0.5, 0.5)	8.4	10.6	24.7	41.7	10.9	3.7
U2 - Beta(0.7, 0.3)	8.7	10.3	23.3	42.7	11.1	3.9
U2 - Beta(1, 1)	9.1	10.3	23.1	43.2	10.3	4
U3 - Beta(0.3, 0.7)	10.5	11.9	23.2	40.8	9.8	3.8
U3 - Beta(0.5, 0.5)	12.7	11.5	22.2	40	9.8	3.8
U3 - Beta(0.7, 0.3)	13.4	10.9	21.5	41.7	8.4	4.1
U3 - Beta(1, 1)	14.4	11.4	20.9	39.2	10.2	3.9
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	4.5	7.2	8.1	5	-
U2 - Beta(0.5, 0.5)	4.5	4.5	7.2	8	5	-
U2 - Beta(0.7, 0.3)	4.6	4.6	7.1	8.1	4.9	-
U2 - Beta(1, 1)	4.6	4.6	7	8	5	-
U3 - Beta(0.3, 0.7)	4.6	4.4	7.2	8	5.1	-
U3 - Beta(0.5, 0.5)	4.4	4.5	7.3	8.1	4.9	-
U3 - Beta(0.7, 0.3)	4.6	4.6	7.1	8.2	4.8	-
U3 - Beta(1, 1)	4.5	4.6	7.1	8.1	5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	2.8	2.3	2.7	1.2	-
U2 - Beta(0.5, 0.5)	0.7	2.9	2.3	2.6	1.2	-
U2 - Beta(0.7, 0.3)	0.7	2.9	2.3	2.6	1.2	-
U2 - Beta(1, 1)	0.7	2.9	2.2	2.7	1.2	-
U3 - Beta(0.3, 0.7)	0.7	2.8	2.3	2.6	1.2	-
U3 - Beta(0.5, 0.5)	0.7	2.9	2.3	2.7	1.2	-
U3 - Beta(0.7, 0.3)	0.7	2.9	2.3	2.7	1.2	-
U3 - Beta(1, 1)	0.7	2.9	2.2	2.7	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1077.8
U2 - Beta(0.5, 0.5)						1077.2
U2 - Beta(0.7, 0.3)						1076.4
U2 - Beta(1, 1)						1076.9
U3 - Beta(0.3, 0.7)						1078.2
U3 - Beta(0.5, 0.5)						1078
U3 - Beta(0.7, 0.3)						1075.4
U3 - Beta(1, 1)						1077

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.2	33.3	27.9	20.6	7.9	7.1
U2 - Beta(0.5, 0.5)	3.5	35.7	27.5	19.1	7.1	7.1
U2 - Beta(0.7, 0.3)	4.3	37	27.2	17.9	6.5	7.1
U2 - Beta(1, 1)	4.8	36.5	26.9	19.2	5.5	7.1
U3 - Beta(0.3, 0.7)	5.7	38.8	26.6	16.8	5	7.1
U3 - Beta(0.5, 0.5)	6.2	40.2	25.6	16.6	4.3	7.1
U3 - Beta(0.7, 0.3)	7.3	41	25	15.9	3.7	7.1
U3 - Beta(1, 1)	8.5	42.2	23.9	14.8	3.5	7.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	6.5	6.8	5.9	4.7	-
U2 - Beta(0.5, 0.5)	4.6	6.8	6.7	5.9	4.5	-
U2 - Beta(0.7, 0.3)	4.7	6.9	6.7	5.9	4.5	-
U2 - Beta(1, 1)	4.7	6.7	6.8	5.9	4.5	-
U3 - Beta(0.3, 0.7)	4.6	6.5	7	5.9	4.6	-
U3 - Beta(0.5, 0.5)	4.6	6.7	6.7	6	4.5	-
U3 - Beta(0.7, 0.3)	4.7	6.7	6.7	6	4.5	-
U3 - Beta(1, 1)	4.7	6.8	6.8	5.8	4.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.5	2.2	2.3	1.7	-
U2 - Beta(0.5, 0.5)	0.7	3.6	2.1	2.2	1.6	-
U2 - Beta(0.7, 0.3)	0.7	3.6	2.1	2.2	1.6	-
U2 - Beta(1, 1)	0.7	3.6	2.1	2.3	1.6	-
U3 - Beta(0.3, 0.7)	0.7	3.5	2.2	2.2	1.7	-
U3 - Beta(0.5, 0.5)	0.7	3.6	2.1	2.2	1.6	-
U3 - Beta(0.7, 0.3)	0.7	3.6	2.1	2.3	1.6	-
U3 - Beta(1, 1)	0.7	3.6	2.1	2.2	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1007.4
U2 - Beta(0.5, 0.5)						1007
U2 - Beta(0.7, 0.3)						1005.9
U2 - Beta(1, 1)						1006.2
U3 - Beta(0.3, 0.7)						1007
U3 - Beta(0.5, 0.5)						1006.9
U3 - Beta(0.7, 0.3)						1005.9
U3 - Beta(1, 1)						1006

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	67.4	8.2	2.1	0.5	0	21.8
U2 - Beta(0.5, 0.5)	66.2	9.7	1.5	0.6	0.1	21.9
U2 - Beta(0.7, 0.3)	66.3	9.7	1.9	0.4	0.1	21.6
U2 - Beta(1, 1)	66.5	9.3	1.8	0.6	0.1	21.7
U3 - Beta(0.3, 0.7)	68.1	8.4	1.2	0.5	0.1	21.7
U3 - Beta(0.5, 0.5)	68.3	8.5	1.4	0.2	0.1	21.5
U3 - Beta(0.7, 0.3)	68	8.5	1.5	0.1	0.1	21.8
U3 - Beta(1, 1)	67.8	8.3	1.6	0.3	0.1	21.9
Mean administrations						
U2 - Beta(0.3, 0.7)	20.1	3.6	1.7	0.6	0.1	-
U2 - Beta(0.5, 0.5)	20.1	3.6	1.7	0.5	0.2	-
U2 - Beta(0.7, 0.3)	20.1	3.6	1.7	0.5	0.2	-
U2 - Beta(1, 1)	20.1	3.6	1.7	0.5	0.2	-
U3 - Beta(0.3, 0.7)	20.2	3.6	1.7	0.5	0.2	-
U3 - Beta(0.5, 0.5)	20.1	3.6	1.7	0.5	0.2	-
U3 - Beta(0.7, 0.3)	20.1	3.6	1.6	0.6	0.2	-
U3 - Beta(1, 1)	20.2	3.6	1.6	0.5	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.6	1.4	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	5.6	1.5	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	5.6	1.5	0.5	0.2	0	-
U2 - Beta(1, 1)	5.6	1.5	0.5	0.2	0	-
U3 - Beta(0.3, 0.7)	5.6	1.4	0.5	0.2	0	-
U3 - Beta(0.5, 0.5)	5.6	1.5	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	5.6	1.5	0.5	0.2	0	-
U3 - Beta(1, 1)	5.6	1.5	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1011.6
U2 - Beta(0.5, 0.5)						1011.5
U2 - Beta(0.7, 0.3)						1011.7
U2 - Beta(1, 1)						1010.4
U3 - Beta(0.3, 0.7)						1010.8
U3 - Beta(0.5, 0.5)						1011.8
U3 - Beta(0.7, 0.3)						1011.7
U3 - Beta(1, 1)						1010.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	29.3	38	17.9	8	1.9	4.9
U2 - Beta(0.5, 0.5)	28.2	39.2	18.3	7.8	1.6	4.9
U2 - Beta(0.7, 0.3)	28.8	38.4	18.3	8.1	1.5	4.9
U2 - Beta(1, 1)	28.6	38.7	17.8	8.3	1.7	4.9
U3 - Beta(0.3, 0.7)	30.7	40.7	15.2	7.2	1.3	4.9
U3 - Beta(0.5, 0.5)	31	41.7	14.7	6.6	1.1	4.9
U3 - Beta(0.7, 0.3)	31.2	41.3	15.4	6.4	0.9	4.8
U3 - Beta(1, 1)	31.2	41.6	13.9	7.5	1	4.8
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	7.3	5.3	3.1	1.6	-
U2 - Beta(0.5, 0.5)	11.7	7.5	5.3	3.1	1.6	-
U2 - Beta(0.7, 0.3)	11.8	7.4	5.3	3.1	1.6	-
U2 - Beta(1, 1)	11.7	7.3	5.3	3.1	1.6	-
U3 - Beta(0.3, 0.7)	11.7	7.4	5.3	3.1	1.6	-
U3 - Beta(0.5, 0.5)	11.8	7.4	5.4	3	1.5	-
U3 - Beta(0.7, 0.3)	11.8	7.4	5.3	3.1	1.5	-
U3 - Beta(1, 1)	11.8	7.3	5.3	3.1	1.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	2.9	1.3	0.9	0.4	-
U2 - Beta(0.5, 0.5)	3.4	3	1.3	0.9	0.4	-
U2 - Beta(0.7, 0.3)	3.4	3	1.3	0.9	0.4	-
U2 - Beta(1, 1)	3.4	2.9	1.3	0.9	0.4	-
U3 - Beta(0.3, 0.7)	3.4	3	1.3	0.9	0.4	-
U3 - Beta(0.5, 0.5)	3.4	3	1.3	0.9	0.4	-
U3 - Beta(0.7, 0.3)	3.4	3	1.3	0.9	0.4	-
U3 - Beta(1, 1)	3.4	2.9	1.3	0.9	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1108.9
U2 - Beta(0.5, 0.5)						1108.5
U2 - Beta(0.7, 0.3)						1108.8
U2 - Beta(1, 1)						1108.6
U3 - Beta(0.3, 0.7)						1109.1
U3 - Beta(0.5, 0.5)						1108.7
U3 - Beta(0.7, 0.3)						1108.5
U3 - Beta(1, 1)						1109.7
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.3	5.5	1.1	0.1	0	66
U2 - Beta(0.5, 0.5)	27.5	5.4	0.9	0.1	0	66.1
U2 - Beta(0.7, 0.3)	27.8	5.2	0.8	0	0	66.2
U2 - Beta(1, 1)	27.3	5.7	0.8	0.1	0	66.1
U3 - Beta(0.3, 0.7)	27.8	5.2	0.9	0	0	66.1
U3 - Beta(0.5, 0.5)	28.8	4.6	0.7	0.1	0	65.8
U3 - Beta(0.7, 0.3)	28.8	4.5	0.5	0.1	0	66.1
U3 - Beta(1, 1)	28.6	4.5	0.5	0.2	0	66.2
Mean administrations						
U2 - Beta(0.3, 0.7)	14.6	2.8	1.1	0.2	0.1	-
U2 - Beta(0.5, 0.5)	14.6	2.8	1.1	0.2	0.1	-
U2 - Beta(0.7, 0.3)	14.7	2.7	1	0.2	0.1	-
U2 - Beta(1, 1)	14.7	2.8	1	0.2	0.1	-
U3 - Beta(0.3, 0.7)	14.6	2.8	1.1	0.2	0.1	-
U3 - Beta(0.5, 0.5)	14.6	2.8	1.1	0.3	0.1	-
U3 - Beta(0.7, 0.3)	14.7	2.8	1	0.2	0.1	-
U3 - Beta(1, 1)	14.7	2.8	1	0.3	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.4	1.4	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	4.4	1.4	0.4	0.1	0	-
U2 - Beta(0.7, 0.3)	4.4	1.4	0.4	0.1	0	-
U2 - Beta(1, 1)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(0.3, 0.7)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(0.7, 0.3)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(1, 1)	4.4	1.4	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						687.4
U2 - Beta(0.5, 0.5)						686.2
U2 - Beta(0.7, 0.3)						686.2
U2 - Beta(1, 1)						685.9
U3 - Beta(0.3, 0.7)						685.6
U3 - Beta(0.5, 0.5)						686.3
U3 - Beta(0.7, 0.3)						686.1
U3 - Beta(1, 1)						685.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.3	13.5	17	11.3	15.9	33
U2 - Beta(0.5, 0.5)	11.2	13.7	15.6	11.1	15.7	32.7
U2 - Beta(0.7, 0.3)	13	12.4	15.4	11.6	14.9	32.7
U2 - Beta(1, 1)	13	13	14.8	10.9	15.3	33
U3 - Beta(0.3, 0.7)	14.6	12.7	15.5	9.8	15	32.4
U3 - Beta(0.5, 0.5)	15.3	12.7	14.4	10.3	14.4	32.9
U3 - Beta(0.7, 0.3)	18.5	12.1	12.3	9.6	14.6	32.9
U3 - Beta(1, 1)	20.2	12	11.5	9	14.3	33
Mean administrations						
U2 - Beta(0.3, 0.7)	7.9	4.5	5.2	3.5	3.5	-
U2 - Beta(0.5, 0.5)	8	4.5	5.2	3.4	3.5	-
U2 - Beta(0.7, 0.3)	8	4.7	5.2	3.5	3.5	-
U2 - Beta(1, 1)	8	4.7	5.1	3.5	3.5	-
U3 - Beta(0.3, 0.7)	7.9	4.5	5.4	3.4	3.5	-
U3 - Beta(0.5, 0.5)	8	4.5	5.2	3.4	3.5	-
U3 - Beta(0.7, 0.3)	8	4.6	5.2	3.4	3.6	-
U3 - Beta(1, 1)	8	4.7	5.1	3.5	3.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	2.6	1.9	1.4	1	-
U2 - Beta(0.5, 0.5)	2.2	2.6	1.8	1.4	1.1	-
U2 - Beta(0.7, 0.3)	2.1	2.6	1.8	1.4	1	-
U2 - Beta(1, 1)	2.1	2.7	1.8	1.4	1.1	-
U3 - Beta(0.3, 0.7)	2.1	2.6	1.9	1.4	1.1	-
U3 - Beta(0.5, 0.5)	2.1	2.6	1.9	1.4	1.1	-
U3 - Beta(0.7, 0.3)	2.1	2.6	1.8	1.4	1.1	-
U3 - Beta(1, 1)	2.1	2.7	1.8	1.4	1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						859.2
U2 - Beta(0.5, 0.5)						859.5
U2 - Beta(0.7, 0.3)						859.3
U2 - Beta(1, 1)						859
U3 - Beta(0.3, 0.7)						859.4
U3 - Beta(0.5, 0.5)						859.4
U3 - Beta(0.7, 0.3)						860.2
U3 - Beta(1, 1)						858.6

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.5	25.3	8.7	2.3	0.7	54.5
U2 - Beta(0.5, 0.5)	8.8	25	9.4	2.2	0.6	54
U2 - Beta(0.7, 0.3)	8.8	26.3	8.3	2	0.4	54.2
U2 - Beta(1, 1)	9.1	25.4	9.2	1.8	0.5	54
U3 - Beta(0.3, 0.7)	9.3	25.5	8.5	1.8	0.6	54.3
U3 - Beta(0.5, 0.5)	9.4	26.1	8.2	1.4	0.6	54.3
U3 - Beta(0.7, 0.3)	9.3	26.7	7.5	1.6	0.5	54.4
U3 - Beta(1, 1)	9.5	26.2	8.1	1.5	0.3	54.4
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	5.3	3.5	1.4	0.4	-
U2 - Beta(0.5, 0.5)	9.8	5.4	3.4	1.4	0.4	-
U2 - Beta(0.7, 0.3)	9.8	5.5	3.3	1.4	0.4	-
U2 - Beta(1, 1)	9.8	5.4	3.3	1.4	0.4	-
U3 - Beta(0.3, 0.7)	9.8	5.3	3.4	1.4	0.4	-
U3 - Beta(0.5, 0.5)	9.8	5.5	3.3	1.4	0.4	-
U3 - Beta(0.7, 0.3)	9.8	5.4	3.3	1.4	0.4	-
U3 - Beta(1, 1)	9.8	5.4	3.4	1.4	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	2.4	1	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2.7	2.4	0.9	0.4	0.1	-
U2 - Beta(0.7, 0.3)	2.7	2.4	0.9	0.5	0.1	-
U2 - Beta(1, 1)	2.7	2.4	0.9	0.5	0.1	-
U3 - Beta(0.3, 0.7)	2.7	2.4	0.9	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.7	2.4	0.9	0.4	0.1	-
U3 - Beta(0.7, 0.3)	2.7	2.4	0.9	0.5	0.1	-
U3 - Beta(1, 1)	2.7	2.4	0.9	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						749.7
U2 - Beta(0.5, 0.5)						750.6
U2 - Beta(0.7, 0.3)						749.6
U2 - Beta(1, 1)						749.8
U3 - Beta(0.3, 0.7)						749.4
U3 - Beta(0.5, 0.5)						748.5
U3 - Beta(0.7, 0.3)						749.2
U3 - Beta(1, 1)						749.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	68.1	16.7	0.8	0	0	14.4
U2 - Beta(0.5, 0.5)	67.9	16.6	1.2	0	0	14.3
U2 - Beta(0.7, 0.3)	66.9	17.5	1.5	0	0	14.1
U2 - Beta(1, 1)	67.4	17.1	1.4	0	0	14.1
U3 - Beta(0.3, 0.7)	69.6	15.2	0.7	0	0	14.5
U3 - Beta(0.5, 0.5)	69	15.8	0.9	0	0	14.3
U3 - Beta(0.7, 0.3)	68.7	16	0.9	0	0	14.4
U3 - Beta(1, 1)	70.2	15.1	0.7	0	0	14
Mean administrations						
U2 - Beta(0.3, 0.7)	19.1	5.5	2.4	0.5	0	-
U2 - Beta(0.5, 0.5)	19.1	5.5	2.4	0.5	0	-
U2 - Beta(0.7, 0.3)	19.2	5.5	2.4	0.5	0	-
U2 - Beta(1, 1)	19.2	5.4	2.4	0.5	0	-
U3 - Beta(0.3, 0.7)	19.1	5.5	2.5	0.5	0	-
U3 - Beta(0.5, 0.5)	19.1	5.5	2.4	0.5	0	-
U3 - Beta(0.7, 0.3)	19.2	5.5	2.4	0.5	0	-
U3 - Beta(1, 1)	19.3	5.4	2.4	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6	2.4	0.8	0.2	0	-
U2 - Beta(0.5, 0.5)	6	2.4	0.8	0.2	0	-
U2 - Beta(0.7, 0.3)	6	2.4	0.8	0.2	0	-
U2 - Beta(1, 1)	6	2.4	0.8	0.2	0	-
U3 - Beta(0.3, 0.7)	6	2.5	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	6	2.5	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	6	2.5	0.8	0.2	0	-
U3 - Beta(1, 1)	6.1	2.4	0.8	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						993.6
U2 - Beta(0.5, 0.5)						994.2
U2 - Beta(0.7, 0.3)						996.2
U2 - Beta(1, 1)						993.4
U3 - Beta(0.3, 0.7)						994
U3 - Beta(0.5, 0.5)						994.8
U3 - Beta(0.7, 0.3)						994.1
U3 - Beta(1, 1)						994.6
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.6	17.5	14.5	36.3	13.3	5.8
U2 - Beta(0.5, 0.5)	14.8	17.1	12.6	37.1	12.9	5.5
U2 - Beta(0.7, 0.3)	16.2	17.1	11.6	36.4	13.1	5.6
U2 - Beta(1, 1)	17	16.8	11.1	36.3	13	5.8
U3 - Beta(0.3, 0.7)	19.1	17.8	10.7	34.6	12	5.8
U3 - Beta(0.5, 0.5)	21.1	17.6	10	33.4	12.1	5.8
U3 - Beta(0.7, 0.3)	22.5	17.7	7.4	34.5	12.2	5.7
U3 - Beta(1, 1)	25	16.7	7.5	32.6	12.5	5.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5	5	6.1	7.4	5.6	-
U2 - Beta(0.5, 0.5)	4.9	5.3	6	7.4	5.5	-
U2 - Beta(0.7, 0.3)	5.1	5.4	5.8	7.3	5.5	-
U2 - Beta(1, 1)	5.1	5.4	5.9	7.3	5.5	-
U3 - Beta(0.3, 0.7)	4.9	5.2	6	7.5	5.6	-
U3 - Beta(0.5, 0.5)	5	5.3	6	7.3	5.5	-
U3 - Beta(0.7, 0.3)	5.1	5.4	5.8	7.3	5.5	-
U3 - Beta(1, 1)	5.1	5.4	5.8	7.4	5.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.1	2	2.8	1.3	-
U2 - Beta(0.5, 0.5)	0.9	3.2	2	2.8	1.3	-
U2 - Beta(0.7, 0.3)	1	3.2	1.9	2.8	1.3	-
U2 - Beta(1, 1)	1	3.2	1.9	2.8	1.3	-
U3 - Beta(0.3, 0.7)	0.9	3.2	1.9	2.8	1.3	-
U3 - Beta(0.5, 0.5)	0.9	3.2	1.9	2.7	1.3	-
U3 - Beta(0.7, 0.3)	1	3.2	1.9	2.7	1.3	-
U3 - Beta(1, 1)	1	3.2	1.9	2.8	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1048.7
U2 - Beta(0.5, 0.5)						1047.3
U2 - Beta(0.7, 0.3)						1046.9
U2 - Beta(1, 1)						1045.7
U3 - Beta(0.3, 0.7)						1048.8
U3 - Beta(0.5, 0.5)						1046.8
U3 - Beta(0.7, 0.3)						1045.8
U3 - Beta(1, 1)						1047.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.6	26.3	10.2	2	0.6	50.3
U2 - Beta(0.5, 0.5)	11.4	26.2	9.7	1.9	0.6	50.2
U2 - Beta(0.7, 0.3)	11.8	26.6	9.2	1.7	0.5	50.2
U2 - Beta(1, 1)	12.1	26.1	9.3	1.9	0.5	50.1
U3 - Beta(0.3, 0.7)	11.8	27.1	9.1	1.6	0.5	49.9
U3 - Beta(0.5, 0.5)	12.1	26.6	9.2	1.4	0.5	50.2
U3 - Beta(0.7, 0.3)	12.3	27.6	8.1	1.5	0.4	50.1
U3 - Beta(1, 1)	12.5	27.2	8.6	1.1	0.5	50.1
Mean administrations						
U2 - Beta(0.3, 0.7)	9.6	5.8	3.7	1.5	0.5	-
U2 - Beta(0.5, 0.5)	9.6	5.8	3.7	1.4	0.5	-
U2 - Beta(0.7, 0.3)	9.6	5.9	3.6	1.5	0.5	-
U2 - Beta(1, 1)	9.6	5.9	3.6	1.5	0.5	-
U3 - Beta(0.3, 0.7)	9.6	5.8	3.7	1.5	0.5	-
U3 - Beta(0.5, 0.5)	9.6	5.8	3.7	1.4	0.5	-
U3 - Beta(0.7, 0.3)	9.6	5.9	3.6	1.5	0.5	-
U3 - Beta(1, 1)	9.6	5.8	3.6	1.5	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	2.7	1.1	0.6	0.2	-
U2 - Beta(0.5, 0.5)	2.7	2.7	1.1	0.6	0.2	-
U2 - Beta(0.7, 0.3)	2.7	2.8	1.1	0.6	0.2	-
U2 - Beta(1, 1)	2.7	2.8	1.1	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.7	2.7	1.1	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.7	2.7	1.1	0.5	0.2	-
U3 - Beta(0.7, 0.3)	2.7	2.8	1.1	0.6	0.2	-
U3 - Beta(1, 1)	2.7	2.8	1.1	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						759.3
U2 - Beta(0.5, 0.5)						757.5
U2 - Beta(0.7, 0.3)						755.8
U2 - Beta(1, 1)						757.8
U3 - Beta(0.3, 0.7)						757.3
U3 - Beta(0.5, 0.5)						756.7
U3 - Beta(0.7, 0.3)						756.8
U3 - Beta(1, 1)						756.7
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.2	17.8	16.8	22	10.8	16.4
U2 - Beta(0.5, 0.5)	18	18.5	14	24.4	8.7	16.4
U2 - Beta(0.7, 0.3)	18.3	17.7	14	23.7	9.5	16.8
U2 - Beta(1, 1)	18.9	17.3	14.6	22.1	10.1	17
U3 - Beta(0.3, 0.7)	23	17.4	12.4	22.4	8.2	16.6
U3 - Beta(0.5, 0.5)	24.8	16.8	11.7	21.8	7.8	17.1
U3 - Beta(0.7, 0.3)	28.6	17.2	9.4	20.8	6.9	17.1
U3 - Beta(1, 1)	29.9	16.6	9.9	20.1	7	16.5
Mean administrations						
U2 - Beta(0.3, 0.7)	6.5	5.7	6.2	5.2	3.9	-
U2 - Beta(0.5, 0.5)	6.7	5.7	6.1	5.2	3.7	-
U2 - Beta(0.7, 0.3)	6.7	5.9	5.9	5.3	3.7	-
U2 - Beta(1, 1)	6.8	5.8	5.8	5.2	3.8	-
U3 - Beta(0.3, 0.7)	6.6	5.6	6.2	5.2	3.9	-
U3 - Beta(0.5, 0.5)	6.6	5.7	6	5.3	3.8	-
U3 - Beta(0.7, 0.3)	6.7	5.8	6	5.1	3.8	-
U3 - Beta(1, 1)	6.8	5.9	5.8	5.2	3.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	3.3	2.2	2	1.2	-
U2 - Beta(0.5, 0.5)	1.6	3.3	2.1	2	1.2	-
U2 - Beta(0.7, 0.3)	1.7	3.3	2.1	2.1	1.1	-
U2 - Beta(1, 1)	1.7	3.3	2	2	1.2	-
U3 - Beta(0.3, 0.7)	1.6	3.2	2.2	2	1.2	-
U3 - Beta(0.5, 0.5)	1.6	3.2	2.1	2.1	1.2	-
U3 - Beta(0.7, 0.3)	1.6	3.3	2.1	2	1.2	-
U3 - Beta(1, 1)	1.7	3.3	2.1	2	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						948.8
U2 - Beta(0.5, 0.5)						947.5
U2 - Beta(0.7, 0.3)						945.5
U2 - Beta(1, 1)						945.4
U3 - Beta(0.3, 0.7)						947.7
U3 - Beta(0.5, 0.5)						947.9
U3 - Beta(0.7, 0.3)						945.9
U3 - Beta(1, 1)						944.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	39.7	28.8	9.6	3	0	18.9
U2 - Beta(0.5, 0.5)	41.7	26.8	9.2	2.9	0.2	19.2
U2 - Beta(0.7, 0.3)	42.5	26.6	9.2	2.5	0.1	19.1
U2 - Beta(1, 1)	44.6	25.7	8.5	2.3	0	18.9
U3 - Beta(0.3, 0.7)	47.2	25.7	6.1	2	0.1	18.9
U3 - Beta(0.5, 0.5)	50.2	24.2	5.1	1.1	0.2	19.2
U3 - Beta(0.7, 0.3)	52.6	22.6	4	1.6	0.1	19.1
U3 - Beta(1, 1)	53.8	21.6	4.2	1.5	0.1	18.8
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	7.4	5.5	2.5	0.8	-
U2 - Beta(0.5, 0.5)	11.3	7.4	5.4	2.5	0.8	-
U2 - Beta(0.7, 0.3)	11.3	7.5	5.3	2.4	0.8	-
U2 - Beta(1, 1)	11.3	7.4	5.4	2.4	0.8	-
U3 - Beta(0.3, 0.7)	11.2	7.4	5.5	2.4	0.9	-
U3 - Beta(0.5, 0.5)	11.2	7.5	5.4	2.4	0.9	-
U3 - Beta(0.7, 0.3)	11.3	7.5	5.3	2.4	0.8	-
U3 - Beta(1, 1)	11.4	7.4	5.3	2.4	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	3.7	2	1.1	0.3	-
U2 - Beta(0.5, 0.5)	3.5	3.7	1.9	1.1	0.3	-
U2 - Beta(0.7, 0.3)	3.5	3.7	1.9	1.1	0.3	-
U2 - Beta(1, 1)	3.5	3.7	1.9	1.1	0.3	-
U3 - Beta(0.3, 0.7)	3.4	3.7	2	1.1	0.4	-
U3 - Beta(0.5, 0.5)	3.4	3.8	1.9	1.1	0.4	-
U3 - Beta(0.7, 0.3)	3.5	3.7	1.9	1.1	0.3	-
U3 - Beta(1, 1)	3.5	3.7	1.9	1.1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						924.1
U2 - Beta(0.5, 0.5)						923.1
U2 - Beta(0.7, 0.3)						922.8
U2 - Beta(1, 1)						923.1
U3 - Beta(0.3, 0.7)						922.9
U3 - Beta(0.5, 0.5)						922.6
U3 - Beta(0.7, 0.3)						922.8
U3 - Beta(1, 1)						921.7
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	28.3	27.8	17.5	15.9	10.4	0.1
U2 - Beta(0.5, 0.5)	27.8	28.1	17.3	15.2	11.5	0.1
U2 - Beta(0.7, 0.3)	26.8	29.7	16.3	15.8	11.3	0.1
U2 - Beta(1, 1)	27.9	29.4	17.8	14.9	9.9	0.1
U3 - Beta(0.3, 0.7)	33.8	31.9	15.3	12.4	6.5	0.1
U3 - Beta(0.5, 0.5)	34.5	31.6	15	11.9	6.9	0.1
U3 - Beta(0.7, 0.3)	34.3	34.1	14.2	11.4	5.9	0.1
U3 - Beta(1, 1)	37.6	32.7	13.9	10.7	5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	5.7	5.5	5.8	7.3	-
U2 - Beta(0.5, 0.5)	5.8	5.9	5.5	5.7	7	-
U2 - Beta(0.7, 0.3)	5.7	5.9	5.5	6	6.9	-
U2 - Beta(1, 1)	5.7	5.9	5.6	5.9	6.9	-
U3 - Beta(0.3, 0.7)	5.7	5.7	5.5	5.8	7.3	-
U3 - Beta(0.5, 0.5)	5.8	5.9	5.5	5.8	7	-
U3 - Beta(0.7, 0.3)	5.7	6	5.5	5.9	6.9	-
U3 - Beta(1, 1)	5.7	5.9	5.5	6	6.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	3.2	1.9	2.6	2.8	-
U2 - Beta(0.5, 0.5)	1.2	3.3	1.9	2.6	2.7	-
U2 - Beta(0.7, 0.3)	1.1	3.3	1.9	2.7	2.6	-
U2 - Beta(1, 1)	1.1	3.3	1.9	2.7	2.7	-
U3 - Beta(0.3, 0.7)	1.1	3.2	1.9	2.6	2.8	-
U3 - Beta(0.5, 0.5)	1.1	3.3	1.9	2.7	2.7	-
U3 - Beta(0.7, 0.3)	1.1	3.3	1.9	2.7	2.7	-
U3 - Beta(1, 1)	1.1	3.3	1.9	2.7	2.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1007.8
U2 - Beta(0.5, 0.5)						1007.2
U2 - Beta(0.7, 0.3)						1006.9
U2 - Beta(1, 1)						1007
U3 - Beta(0.3, 0.7)						1007.5
U3 - Beta(0.5, 0.5)						1006.8
U3 - Beta(0.7, 0.3)						1007
U3 - Beta(1, 1)						1007.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31.8	54.7	12.3	0.7	0	0.5
U2 - Beta(0.5, 0.5)	31.5	53.9	13.1	1	0	0.5
U2 - Beta(0.7, 0.3)	31.9	53.5	12.9	1.2	0	0.5
U2 - Beta(1, 1)	31.6	53.7	13.3	0.9	0	0.5
U3 - Beta(0.3, 0.7)	35.8	53.2	9.7	0.8	0	0.5
U3 - Beta(0.5, 0.5)	34.6	53.9	10.3	0.7	0	0.5
U3 - Beta(0.7, 0.3)	34.2	54.3	10	1	0	0.5
U3 - Beta(1, 1)	35.5	53.6	9.8	0.6	0	0.5
Mean administrations						
U2 - Beta(0.3, 0.7)	11	10.5	6.5	1.6	0.2	-
U2 - Beta(0.5, 0.5)	11.1	10.5	6.5	1.5	0.2	-
U2 - Beta(0.7, 0.3)	11.1	10.6	6.5	1.5	0.2	-
U2 - Beta(1, 1)	11.1	10.5	6.5	1.6	0.2	-
U3 - Beta(0.3, 0.7)	11.1	10.6	6.5	1.5	0.2	-
U3 - Beta(0.5, 0.5)	11.1	10.5	6.5	1.6	0.2	-
U3 - Beta(0.7, 0.3)	11.2	10.5	6.5	1.5	0.2	-
U3 - Beta(1, 1)	11.1	10.6	6.5	1.5	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	3.6	1.5	0.5	0	-
U2 - Beta(0.5, 0.5)	2.9	3.6	1.5	0.5	0.1	-
U2 - Beta(0.7, 0.3)	2.8	3.6	1.5	0.5	0	-
U2 - Beta(1, 1)	2.8	3.6	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	2.9	3.6	1.5	0.5	0	-
U3 - Beta(0.5, 0.5)	2.9	3.6	1.5	0.5	0.1	-
U3 - Beta(0.7, 0.3)	2.9	3.6	1.5	0.4	0.1	-
U3 - Beta(1, 1)	2.9	3.6	1.5	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1183.9
U2 - Beta(0.5, 0.5)						1183.5
U2 - Beta(0.7, 0.3)						1182.9
U2 - Beta(1, 1)						1183.9
U3 - Beta(0.3, 0.7)						1183.9
U3 - Beta(0.5, 0.5)						1183.1
U3 - Beta(0.7, 0.3)						1182.9
U3 - Beta(1, 1)						1183.1
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.4	11.4	12	18.5	48.9	0.8
U2 - Beta(0.5, 0.5)	9.7	11.9	11.5	17.9	48	1
U2 - Beta(0.7, 0.3)	10.8	13	10.6	17.4	47.3	0.9
U2 - Beta(1, 1)	10.8	12.8	10.4	17	48.2	0.8
U3 - Beta(0.3, 0.7)	13.7	13.2	11.2	15.9	45.1	0.9
U3 - Beta(0.5, 0.5)	14.5	14.3	11.2	16.1	43	0.9
U3 - Beta(0.7, 0.3)	16.7	13.6	10.9	14.9	43	0.9
U3 - Beta(1, 1)	18.7	14.2	10.2	13.8	42.1	1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.7	3.9	4.8	5.7	11.8	-
U2 - Beta(0.5, 0.5)	3.7	4	4.9	5.7	11.5	-
U2 - Beta(0.7, 0.3)	3.7	4	4.9	5.7	11.5	-
U2 - Beta(1, 1)	3.7	4.1	4.9	5.7	11.5	-
U3 - Beta(0.3, 0.7)	3.7	3.9	4.7	5.7	11.9	-
U3 - Beta(0.5, 0.5)	3.7	4	4.9	5.7	11.6	-
U3 - Beta(0.7, 0.3)	3.8	4	4.9	5.8	11.4	-
U3 - Beta(1, 1)	3.8	4.1	4.9	5.7	11.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.7	1.6	2.5	3.3	-
U2 - Beta(0.5, 0.5)	0.3	2.8	1.6	2.6	3.2	-
U2 - Beta(0.7, 0.3)	0.4	2.8	1.6	2.6	3.2	-
U2 - Beta(1, 1)	0.3	2.8	1.6	2.6	3.2	-
U3 - Beta(0.3, 0.7)	0.3	2.7	1.5	2.6	3.3	-
U3 - Beta(0.5, 0.5)	0.3	2.8	1.6	2.6	3.2	-
U3 - Beta(0.7, 0.3)	0.4	2.8	1.6	2.6	3.2	-
U3 - Beta(1, 1)	0.4	2.8	1.6	2.5	3.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1069.6
U2 - Beta(0.5, 0.5)						1065.9
U2 - Beta(0.7, 0.3)						1064.8
U2 - Beta(1, 1)						1065.2
U3 - Beta(0.3, 0.7)						1069.6
U3 - Beta(0.5, 0.5)						1066.6
U3 - Beta(0.7, 0.3)						1064
U3 - Beta(1, 1)						1064.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.3	48.9	11.4	0.5	0	16.9
U2 - Beta(0.5, 0.5)	22.2	48.8	11.2	0.5	0	17.3
U2 - Beta(0.7, 0.3)	22.8	49.1	10.3	0.6	0	17.2
U2 - Beta(1, 1)	23.9	48.1	10.2	0.5	0	17.3
U3 - Beta(0.3, 0.7)	23.3	49.3	9.7	0.3	0	17.4
U3 - Beta(0.5, 0.5)	24.8	48.2	9.4	0.4	0	17.2
U3 - Beta(0.7, 0.3)	25.5	47.2	9.3	0.9	0	17.1
U3 - Beta(1, 1)	25.8	48.5	7.8	0.6	0	17.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	10.5	6.2	1.3	0.1	-
U2 - Beta(0.5, 0.5)	9	10.6	6	1.3	0.2	-
U2 - Beta(0.7, 0.3)	9	10.7	5.9	1.3	0.1	-
U2 - Beta(1, 1)	9	10.7	5.9	1.3	0.1	-
U3 - Beta(0.3, 0.7)	8.9	10.5	6.2	1.3	0.2	-
U3 - Beta(0.5, 0.5)	9	10.6	6	1.3	0.2	-
U3 - Beta(0.7, 0.3)	9	10.7	5.8	1.3	0.2	-
U3 - Beta(1, 1)	9	10.7	5.9	1.3	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	4.5	1.8	0.5	0	-
U2 - Beta(0.5, 0.5)	2.5	4.6	1.7	0.5	0	-
U2 - Beta(0.7, 0.3)	2.5	4.6	1.7	0.5	0	-
U2 - Beta(1, 1)	2.5	4.6	1.7	0.5	0	-
U3 - Beta(0.3, 0.7)	2.4	4.5	1.8	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.5	4.6	1.7	0.5	0	-
U3 - Beta(0.7, 0.3)	2.5	4.6	1.7	0.5	0	-
U3 - Beta(1, 1)	2.5	4.6	1.7	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						976.9
U2 - Beta(0.5, 0.5)						974.7
U2 - Beta(0.7, 0.3)						975
U2 - Beta(1, 1)						975
U3 - Beta(0.3, 0.7)						976.5
U3 - Beta(0.5, 0.5)						976.4
U3 - Beta(0.7, 0.3)						974.8
U3 - Beta(1, 1)						975.3

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.1	48.2	27.2	5.4	0.3	7.8
U2 - Beta(0.5, 0.5)	10.6	48.9	26.6	6.1	0.1	7.7
U2 - Beta(0.7, 0.3)	11	48.7	25.4	6.9	0.2	7.8
U2 - Beta(1, 1)	13.2	48.7	24.4	5.8	0.3	7.6
U3 - Beta(0.3, 0.7)	14	52.3	22.4	3.6	0.2	7.5
U3 - Beta(0.5, 0.5)	14.8	49.7	22.7	4.7	0.4	7.7
U3 - Beta(0.7, 0.3)	16.8	49.6	20.7	4.7	0.4	7.8
U3 - Beta(1, 1)	17.9	50	19.4	4.6	0.3	7.8
Mean administrations						
U2 - Beta(0.3, 0.7)	5.4	9.3	8.6	3.8	1.4	-
U2 - Beta(0.5, 0.5)	5.5	9.5	8.5	3.7	1.3	-
U2 - Beta(0.7, 0.3)	5.5	9.7	8.3	3.7	1.4	-
U2 - Beta(1, 1)	5.5	9.6	8.4	3.7	1.4	-
U3 - Beta(0.3, 0.7)	5.4	9.4	8.6	3.7	1.4	-
U3 - Beta(0.5, 0.5)	5.5	9.5	8.4	3.7	1.4	-
U3 - Beta(0.7, 0.3)	5.5	9.7	8.2	3.7	1.4	-
U3 - Beta(1, 1)	5.5	9.6	8.3	3.7	1.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	4.5	2.7	1.4	0.5	-
U2 - Beta(0.5, 0.5)	1.1	4.6	2.6	1.4	0.5	-
U2 - Beta(0.7, 0.3)	1.1	4.6	2.5	1.4	0.5	-
U2 - Beta(1, 1)	1.1	4.6	2.6	1.4	0.5	-
U3 - Beta(0.3, 0.7)	1.1	4.5	2.7	1.4	0.5	-
U3 - Beta(0.5, 0.5)	1.1	4.6	2.6	1.4	0.5	-
U3 - Beta(0.7, 0.3)	1.1	4.7	2.5	1.4	0.5	-
U3 - Beta(1, 1)	1.1	4.6	2.5	1.4	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1005.7
U2 - Beta(0.5, 0.5)						1004.6
U2 - Beta(0.7, 0.3)						1004.5
U2 - Beta(1, 1)						1004.6
U3 - Beta(0.3, 0.7)						1005.9
U3 - Beta(0.5, 0.5)						1004.9
U3 - Beta(0.7, 0.3)						1004.1
U3 - Beta(1, 1)						1004.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	47.1	0.3	0	0	0	52.6
U2 - Beta(0.5, 0.5)	47.6	0.4	0	0	0	52
U2 - Beta(0.7, 0.3)	47.4	0.5	0	0	0	52.1
U2 - Beta(1, 1)	47.1	0.5	0	0	0	52.4
U3 - Beta(0.3, 0.7)	47.5	0.3	0	0	0	52.2
U3 - Beta(0.5, 0.5)	47.3	0.4	0	0	0	52.3
U3 - Beta(0.7, 0.3)	47.2	0.5	0	0	0	52.3
U3 - Beta(1, 1)	47.6	0.5	0	0	0	51.9
Mean administrations						
U2 - Beta(0.3, 0.7)	19.6	1.5	0.3	0	0	-
U2 - Beta(0.5, 0.5)	19.7	1.6	0.3	0	0	-
U2 - Beta(0.7, 0.3)	19.7	1.6	0.3	0	0	-
U2 - Beta(1, 1)	19.7	1.6	0.3	0	0	-
U3 - Beta(0.3, 0.7)	19.7	1.5	0.3	0	0	-
U3 - Beta(0.5, 0.5)	19.7	1.6	0.3	0	0	-
U3 - Beta(0.7, 0.3)	19.7	1.6	0.3	0	0	-
U3 - Beta(1, 1)	19.7	1.6	0.3	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4	0.5	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4	0.5	0.1	0	0	-
U2 - Beta(0.7, 0.3)	4	0.5	0.1	0	0	-
U2 - Beta(1, 1)	4	0.5	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4	0.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4	0.5	0.1	0	0	-
U3 - Beta(0.7, 0.3)	4	0.5	0.1	0	0	-
U3 - Beta(1, 1)	4	0.5	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						932.6
U2 - Beta(0.5, 0.5)						933.8
U2 - Beta(0.7, 0.3)						936.6
U2 - Beta(1, 1)						935.5
U3 - Beta(0.3, 0.7)						934
U3 - Beta(0.5, 0.5)						934.4
U3 - Beta(0.7, 0.3)						934.2
U3 - Beta(1, 1)						935.4
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.3	54.7	7.4	0.6	0	11
U2 - Beta(0.5, 0.5)	26.4	53	8.9	0.5	0	11.2
U2 - Beta(0.7, 0.3)	25.9	53.8	8.6	0.6	0	11.1
U2 - Beta(1, 1)	26.2	52.3	9.9	0.4	0	11.2
U3 - Beta(0.3, 0.7)	29.3	53.4	6.1	0.3	0	10.9
U3 - Beta(0.5, 0.5)	29.6	52.3	6.9	0.4	0	10.8
U3 - Beta(0.7, 0.3)	30.4	50.9	6.9	0.6	0	11.2
U3 - Beta(1, 1)	30.9	50.2	7.6	0.4	0	10.9
Mean administrations						
U2 - Beta(0.3, 0.7)	8.7	13	5.2	1	0.1	-
U2 - Beta(0.5, 0.5)	8.8	13	5.2	1	0.1	-
U2 - Beta(0.7, 0.3)	8.9	12.9	5.2	1	0.1	-
U2 - Beta(1, 1)	8.9	12.9	5.2	1	0.1	-
U3 - Beta(0.3, 0.7)	8.6	13.1	5.3	1	0.1	-
U3 - Beta(0.5, 0.5)	8.7	13	5.3	1	0.1	-
U3 - Beta(0.7, 0.3)	8.9	12.8	5.3	1	0.1	-
U3 - Beta(1, 1)	8.9	12.9	5.1	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	5.6	1.5	0.4	0	-
U2 - Beta(0.5, 0.5)	2.3	5.6	1.6	0.4	0	-
U2 - Beta(0.7, 0.3)	2.3	5.6	1.5	0.4	0	-
U2 - Beta(1, 1)	2.4	5.6	1.5	0.4	0	-
U3 - Beta(0.3, 0.7)	2.2	5.6	1.6	0.4	0	-
U3 - Beta(0.5, 0.5)	2.3	5.6	1.5	0.4	0	-
U3 - Beta(0.7, 0.3)	2.3	5.6	1.6	0.4	0	-
U3 - Beta(1, 1)	2.4	5.6	1.5	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						994.5
U2 - Beta(0.5, 0.5)						994.8
U2 - Beta(0.7, 0.3)						995.1
U2 - Beta(1, 1)						993.5
U3 - Beta(0.3, 0.7)						995.7
U3 - Beta(0.5, 0.5)						995.4
U3 - Beta(0.7, 0.3)						993.2
U3 - Beta(1, 1)						993.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6	1.5	0	0	0	92.5
U2 - Beta(0.5, 0.5)	6.3	1.6	0	0	0	92.1
U2 - Beta(0.7, 0.3)	6.3	1.7	0	0	0	92
U2 - Beta(1, 1)	6.1	1.9	0	0	0	92
U3 - Beta(0.3, 0.7)	6.1	1.6	0	0	0	92.3
U3 - Beta(0.5, 0.5)	6.3	1.6	0	0	0	92.1
U3 - Beta(0.7, 0.3)	6.4	1.5	0	0	0	92.1
U3 - Beta(1, 1)	6.3	1.5	0	0	0	92.2
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	1.7	0.2	0	0	-
U2 - Beta(0.5, 0.5)	11.8	1.7	0.2	0	0	-
U2 - Beta(0.7, 0.3)	11.7	1.7	0.2	0	0	-
U2 - Beta(1, 1)	11.8	1.7	0.2	0	0	-
U3 - Beta(0.3, 0.7)	11.8	1.7	0.2	0	0	-
U3 - Beta(0.5, 0.5)	11.8	1.7	0.2	0	0	-
U3 - Beta(0.7, 0.3)	11.8	1.7	0.2	0	0	-
U3 - Beta(1, 1)	11.7	1.7	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	1	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.4	1	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.4	1	0.1	0	0	-
U2 - Beta(1, 1)	3.4	1	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.4	1	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.4	1	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.4	1	0.1	0	0	-
U3 - Beta(1, 1)	3.4	1	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						497.6
U2 - Beta(0.5, 0.5)						498.7
U2 - Beta(0.7, 0.3)						497.5
U2 - Beta(1, 1)						498.2
U3 - Beta(0.3, 0.7)						499.1
U3 - Beta(0.5, 0.5)						498.3
U3 - Beta(0.7, 0.3)						498.6
U3 - Beta(1, 1)						497.7
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.6	1.3	3.4	45.5	49	0.2
U2 - Beta(0.5, 0.5)	0.6	1.7	4.3	47.6	45.7	0.1
U2 - Beta(0.7, 0.3)	0.9	1.9	3.9	48.1	45.1	0.1
U2 - Beta(1, 1)	1	1.7	3.9	47	46.3	0.1
U3 - Beta(0.3, 0.7)	1.3	2.2	4.2	47.9	44.2	0.2
U3 - Beta(0.5, 0.5)	1.7	2.4	5	50.5	40.3	0.1
U3 - Beta(0.7, 0.3)	1.8	3.2	5.1	49.6	40.2	0.1
U3 - Beta(1, 1)	2	3	5.6	49.7	39.6	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.2	3.1	3.6	8.2	11.9	-
U2 - Beta(0.5, 0.5)	3.2	3.1	3.6	8.3	11.8	-
U2 - Beta(0.7, 0.3)	3.2	3.2	3.7	8.7	11.3	-
U2 - Beta(1, 1)	3.2	3.2	3.6	8.3	11.7	-
U3 - Beta(0.3, 0.7)	3.2	3.1	3.6	8.1	12	-
U3 - Beta(0.5, 0.5)	3.2	3.1	3.6	8.3	11.7	-
U3 - Beta(0.7, 0.3)	3.2	3.1	3.7	8.5	11.4	-
U3 - Beta(1, 1)	3.2	3.1	3.6	8.4	11.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U2 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.2	3.2	-
U2 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.7, 0.3)	0.1	2.4	1.1	3.3	3.1	-
U3 - Beta(1, 1)	0.1	2.4	1.1	3.3	3.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1102
U2 - Beta(0.5, 0.5)						1101.3
U2 - Beta(0.7, 0.3)						1099.9
U2 - Beta(1, 1)						1101.6
U3 - Beta(0.3, 0.7)						1101.7
U3 - Beta(0.5, 0.5)						1101.1
U3 - Beta(0.7, 0.3)						1099.9
U3 - Beta(1, 1)						1101.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	34.6	35.9	24.5	2.9	0	2.1
U2 - Beta(0.5, 0.5)	36.2	33.5	24.9	3.3	0	2.1
U2 - Beta(0.7, 0.3)	34.9	35.7	23.5	3.8	0	2.1
U2 - Beta(1, 1)	35.7	34.5	24.5	3.1	0	2.2
U3 - Beta(0.3, 0.7)	39.7	34.7	21.1	2.4	0	2.1
U3 - Beta(0.5, 0.5)	40.4	34.2	20.7	2.7	0	2
U3 - Beta(0.7, 0.3)	39.7	34.4	20.2	3.6	0	2.1
U3 - Beta(1, 1)	40.6	34.2	20.6	2.6	0	2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	8.8	8.7	2	0.2	-
U2 - Beta(0.5, 0.5)	10.1	8.7	8.6	1.9	0.3	-
U2 - Beta(0.7, 0.3)	10.2	8.9	8.4	1.9	0.2	-
U2 - Beta(1, 1)	10.2	8.9	8.4	1.9	0.3	-
U3 - Beta(0.3, 0.7)	10.1	8.7	8.8	1.9	0.2	-
U3 - Beta(0.5, 0.5)	10.1	8.8	8.6	1.9	0.2	-
U3 - Beta(0.7, 0.3)	10.2	8.8	8.4	1.9	0.2	-
U3 - Beta(1, 1)	10.3	8.8	8.5	1.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	3.8	2.4	0.7	0.1	-
U2 - Beta(0.5, 0.5)	2.9	3.8	2.4	0.7	0.1	-
U2 - Beta(0.7, 0.3)	2.9	3.9	2.3	0.7	0.1	-
U2 - Beta(1, 1)	2.9	3.9	2.3	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.8	3.8	2.4	0.7	0.1	-
U3 - Beta(0.5, 0.5)	2.8	3.8	2.4	0.7	0.1	-
U3 - Beta(0.7, 0.3)	2.9	3.9	2.3	0.7	0.1	-
U3 - Beta(1, 1)	2.9	3.8	2.3	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1096.8
U2 - Beta(0.5, 0.5)						1095.2
U2 - Beta(0.7, 0.3)						1094
U2 - Beta(1, 1)						1094.6
U3 - Beta(0.3, 0.7)						1096.8
U3 - Beta(0.5, 0.5)						1094.7
U3 - Beta(0.7, 0.3)						1094
U3 - Beta(1, 1)						1093.7
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.8	30.1	20.8	2.6	0	19.7
U2 - Beta(0.5, 0.5)	27.3	29.6	20.9	2.5	0	19.7
U2 - Beta(0.7, 0.3)	28.7	27.9	20.8	2.5	0	20.1
U2 - Beta(1, 1)	29	28	20.4	2.5	0	20.1
U3 - Beta(0.3, 0.7)	30	29.4	18.5	2	0	20.1
U3 - Beta(0.5, 0.5)	30.3	29.3	18.2	2	0	20.2
U3 - Beta(0.7, 0.3)	32.2	27.8	17.6	2.1	0	20.3
U3 - Beta(1, 1)	32.3	28.1	17.4	2	0	20.2
Mean administrations						
U2 - Beta(0.3, 0.7)	10.7	7.2	6.9	1.9	0.2	-
U2 - Beta(0.5, 0.5)	10.7	7.3	6.9	1.8	0.2	-
U2 - Beta(0.7, 0.3)	10.9	7.2	6.7	1.8	0.2	-
U2 - Beta(1, 1)	10.8	7.2	6.7	1.9	0.2	-
U3 - Beta(0.3, 0.7)	10.7	7.2	6.9	1.9	0.2	-
U3 - Beta(0.5, 0.5)	10.8	7.3	6.8	1.8	0.3	-
U3 - Beta(0.7, 0.3)	10.8	7.2	6.7	1.9	0.2	-
U3 - Beta(1, 1)	10.8	7.3	6.7	1.8	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3.4	2.1	0.7	0.1	-
U2 - Beta(0.5, 0.5)	3.2	3.4	2	0.7	0	-
U2 - Beta(0.7, 0.3)	3.2	3.4	2	0.7	0.1	-
U2 - Beta(1, 1)	3.2	3.4	2	0.7	0.1	-
U3 - Beta(0.3, 0.7)	3.2	3.4	2.1	0.7	0.1	-
U3 - Beta(0.5, 0.5)	3.2	3.4	2	0.7	0.1	-
U3 - Beta(0.7, 0.3)	3.3	3.4	2	0.7	0	-
U3 - Beta(1, 1)	3.2	3.4	2	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						965.3
U2 - Beta(0.5, 0.5)						964.2
U2 - Beta(0.7, 0.3)						963
U2 - Beta(1, 1)						963.9
U3 - Beta(0.3, 0.7)						965
U3 - Beta(0.5, 0.5)						963.5
U3 - Beta(0.7, 0.3)						962.8
U3 - Beta(1, 1)						962.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.1	27.6	36.7	4.3	0.1	12.2
U2 - Beta(0.5, 0.5)	19.1	27.6	37.8	3.5	0	12
U2 - Beta(0.7, 0.3)	19.7	28.8	35.4	4.3	0	11.8
U2 - Beta(1, 1)	20	27.9	36.2	4.1	0	11.8
U3 - Beta(0.3, 0.7)	21.9	27.2	35.1	3.6	0	12.2
U3 - Beta(0.5, 0.5)	23	28.7	33	3.4	0	11.9
U3 - Beta(0.7, 0.3)	23.4	28.3	32.4	4	0	11.9
U3 - Beta(1, 1)	23.6	28.7	32.1	3.9	0	11.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	7	9.1	3.1	0.8	-
U2 - Beta(0.5, 0.5)	8.2	7.1	9	3.1	0.8	-
U2 - Beta(0.7, 0.3)	8.3	7.1	9	3	0.7	-
U2 - Beta(1, 1)	8.2	7	9	3.1	0.8	-
U3 - Beta(0.3, 0.7)	8.1	6.9	9.3	3.1	0.8	-
U3 - Beta(0.5, 0.5)	8.2	7.1	9	3.1	0.8	-
U3 - Beta(0.7, 0.3)	8.2	7.1	9	3.1	0.8	-
U3 - Beta(1, 1)	8.2	7	9.1	3.1	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	3.4	2.7	1	0.2	-
U2 - Beta(0.5, 0.5)	2.2	3.4	2.6	1	0.2	-
U2 - Beta(0.7, 0.3)	2.3	3.4	2.6	0.9	0.2	-
U2 - Beta(1, 1)	2.3	3.4	2.6	1	0.2	-
U3 - Beta(0.3, 0.7)	2.2	3.4	2.7	1	0.2	-
U3 - Beta(0.5, 0.5)	2.2	3.4	2.6	1	0.2	-
U3 - Beta(0.7, 0.3)	2.3	3.4	2.6	1	0.2	-
U3 - Beta(1, 1)	2.3	3.4	2.6	1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1028.5
U2 - Beta(0.5, 0.5)						1030
U2 - Beta(0.7, 0.3)						1028.7
U2 - Beta(1, 1)						1028.9
U3 - Beta(0.3, 0.7)						1030.6
U3 - Beta(0.5, 0.5)						1029.8
U3 - Beta(0.7, 0.3)						1028.3
U3 - Beta(1, 1)						1029.9
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.5	27.3	1.5	0	0	52.7
U2 - Beta(0.5, 0.5)	18.2	28	1.2	0	0	52.6
U2 - Beta(0.7, 0.3)	18.5	27	1.5	0	0	53
U2 - Beta(1, 1)	18	28.1	1.5	0	0	52.4
U3 - Beta(0.3, 0.7)	19.1	26.7	1.3	0	0	52.9
U3 - Beta(0.5, 0.5)	19	27.7	1	0	0	52.3
U3 - Beta(0.7, 0.3)	19.5	27	1.3	0	0	52.2
U3 - Beta(1, 1)	20.1	26.8	1.2	0	0	51.9
Mean administrations						
U2 - Beta(0.3, 0.7)	10.3	7.4	2.9	0.4	0	-
U2 - Beta(0.5, 0.5)	10.2	7.4	3	0.4	0	-
U2 - Beta(0.7, 0.3)	10.3	7.4	2.9	0.5	0	-
U2 - Beta(1, 1)	10.3	7.4	2.9	0.5	0	-
U3 - Beta(0.3, 0.7)	10.2	7.4	2.9	0.5	0	-
U3 - Beta(0.5, 0.5)	10.2	7.5	2.9	0.4	0	-
U3 - Beta(0.7, 0.3)	10.3	7.4	2.9	0.4	0	-
U3 - Beta(1, 1)	10.3	7.4	2.9	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	3.2	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	2.9	3.2	0.9	0.2	0	-
U2 - Beta(0.7, 0.3)	2.9	3.2	0.9	0.2	0	-
U2 - Beta(1, 1)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(0.3, 0.7)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(0.5, 0.5)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(0.7, 0.3)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(1, 1)	2.9	3.2	0.9	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						761.3
U2 - Beta(0.5, 0.5)						760.5
U2 - Beta(0.7, 0.3)						761.3
U2 - Beta(1, 1)						761.4
U3 - Beta(0.3, 0.7)						760.7
U3 - Beta(0.5, 0.5)						762.4
U3 - Beta(0.7, 0.3)						762.7
U3 - Beta(1, 1)						760.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	75.6	19.7	1.3	0	0	3.4
U2 - Beta(0.5, 0.5)	74.2	20.8	1.7	0	0	3.3
U2 - Beta(0.7, 0.3)	73.4	21.6	1.8	0	0	3.2
U2 - Beta(1, 1)	75.8	19.5	1.4	0	0	3.3
U3 - Beta(0.3, 0.7)	77.6	18.4	0.7	0	0	3.3
U3 - Beta(0.5, 0.5)	76.7	18.8	1.1	0	0	3.4
U3 - Beta(0.7, 0.3)	76.3	19.2	1.3	0	0	3.2
U3 - Beta(1, 1)	76.6	19.3	0.8	0	0	3.3
Mean administrations						
U2 - Beta(0.3, 0.7)	19.4	7	2.7	0.3	0	-
U2 - Beta(0.5, 0.5)	19.5	7	2.6	0.3	0	-
U2 - Beta(0.7, 0.3)	19.6	6.9	2.6	0.3	0	-
U2 - Beta(1, 1)	19.5	6.9	2.6	0.3	0	-
U3 - Beta(0.3, 0.7)	19.4	7	2.7	0.3	0	-
U3 - Beta(0.5, 0.5)	19.4	7	2.7	0.3	0	-
U3 - Beta(0.7, 0.3)	19.5	7	2.6	0.3	0	-
U3 - Beta(1, 1)	19.5	6.9	2.6	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.1	2.3	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	5.1	2.3	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	5.1	2.3	0.7	0.1	0	-
U2 - Beta(1, 1)	5.1	2.3	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	5.1	2.3	0.8	0.1	0	-
U3 - Beta(0.5, 0.5)	5.1	2.3	0.8	0.1	0	-
U3 - Beta(0.7, 0.3)	5.1	2.3	0.7	0.1	0	-
U3 - Beta(1, 1)	5.1	2.3	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1165.6
U2 - Beta(0.5, 0.5)						1165.7
U2 - Beta(0.7, 0.3)						1165.2
U2 - Beta(1, 1)						1165.7
U3 - Beta(0.3, 0.7)						1166.6
U3 - Beta(0.5, 0.5)						1164.9
U3 - Beta(0.7, 0.3)						1165.4
U3 - Beta(1, 1)						1165.8
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.1	16.7	19.5	36.9	6.9	1.9
U2 - Beta(0.5, 0.5)	17.9	18.1	18.6	36.6	6.9	1.9
U2 - Beta(0.7, 0.3)	20.3	18.6	16.9	34.8	7.5	1.9
U2 - Beta(1, 1)	20.2	18.4	15.9	36.9	6.7	1.9
U3 - Beta(0.3, 0.7)	23.7	18.1	16	33.6	6.7	1.9
U3 - Beta(0.5, 0.5)	23.9	20.4	14.1	33.9	5.7	2
U3 - Beta(0.7, 0.3)	27.7	20.3	14.6	30.1	5.4	1.9
U3 - Beta(1, 1)	28.3	20.9	12.5	30.5	5.9	1.9
Mean administrations						
U2 - Beta(0.3, 0.7)	5.3	4.9	6.3	8.2	5	-
U2 - Beta(0.5, 0.5)	5.3	4.9	6.4	8.2	5	-
U2 - Beta(0.7, 0.3)	5.4	5	6.4	8	5	-
U2 - Beta(1, 1)	5.4	5.1	6.3	8	5	-
U3 - Beta(0.3, 0.7)	5.3	4.9	6.3	8.2	5	-
U3 - Beta(0.5, 0.5)	5.3	5	6.3	8.1	5	-
U3 - Beta(0.7, 0.3)	5.4	5	6.4	7.9	4.9	-
U3 - Beta(1, 1)	5.4	5	6.2	8.1	5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	2.9	2	2.8	1.3	-
U2 - Beta(0.5, 0.5)	1	2.9	2.1	2.8	1.3	-
U2 - Beta(0.7, 0.3)	1.1	2.9	2.1	2.8	1.3	-
U2 - Beta(1, 1)	1	3	2	2.8	1.3	-
U3 - Beta(0.3, 0.7)	1	2.9	2	2.9	1.3	-
U3 - Beta(0.5, 0.5)	1	2.9	2	2.8	1.3	-
U3 - Beta(0.7, 0.3)	1.1	2.9	2	2.8	1.3	-
U3 - Beta(1, 1)	1.1	2.9	2	2.8	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1081
U2 - Beta(0.5, 0.5)						1081.9
U2 - Beta(0.7, 0.3)						1080.1
U2 - Beta(1, 1)						1080.8
U3 - Beta(0.3, 0.7)						1081.9
U3 - Beta(0.5, 0.5)						1080.9
U3 - Beta(0.7, 0.3)						1080.7
U3 - Beta(1, 1)						1081.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5	35	31	23.7	5.1	0.2
U2 - Beta(0.5, 0.5)	5.5	36.2	30.4	22.2	5.5	0.2
U2 - Beta(0.7, 0.3)	4.9	37.2	28.5	23.1	6.1	0.2
U2 - Beta(1, 1)	5.7	35.6	30.8	22.7	5	0.2
U3 - Beta(0.3, 0.7)	6.9	39.6	29.5	20.7	3.1	0.2
U3 - Beta(0.5, 0.5)	7.5	41.6	28.5	18.2	4	0.2
U3 - Beta(0.7, 0.3)	7.9	43.7	25.6	18.8	3.8	0.2
U3 - Beta(1, 1)	8.5	42.2	27.8	18.5	2.8	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.1	5.7	6.6	8	5.5	-
U2 - Beta(0.5, 0.5)	4.2	5.8	6.6	8	5.5	-
U2 - Beta(0.7, 0.3)	4.2	5.9	6.6	7.9	5.3	-
U2 - Beta(1, 1)	4.2	5.7	6.5	8	5.5	-
U3 - Beta(0.3, 0.7)	4.1	5.7	6.5	8	5.5	-
U3 - Beta(0.5, 0.5)	4.1	5.8	6.6	8	5.5	-
U3 - Beta(0.7, 0.3)	4.2	5.9	6.5	7.9	5.4	-
U3 - Beta(1, 1)	4.2	5.8	6.5	8	5.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.9	1.7	2	1.4	-
U2 - Beta(0.5, 0.5)	0.5	2.9	1.7	2.1	1.4	-
U2 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.4	-
U2 - Beta(1, 1)	0.5	2.9	1.7	2	1.4	-
U3 - Beta(0.3, 0.7)	0.5	2.8	1.7	2.1	1.4	-
U3 - Beta(0.5, 0.5)	0.5	2.9	1.7	2	1.4	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.4	-
U3 - Beta(1, 1)	0.5	2.9	1.7	2.1	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1183.9
U2 - Beta(0.5, 0.5)						1182.8
U2 - Beta(0.7, 0.3)						1181.3
U2 - Beta(1, 1)						1183.8
U3 - Beta(0.3, 0.7)						1184.6
U3 - Beta(0.5, 0.5)						1183.7
U3 - Beta(0.7, 0.3)						1182.4
U3 - Beta(1, 1)						1184.7
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.6	13.3	35.2	30.5	10.7	0.7
U2 - Beta(0.5, 0.5)	9.5	14.2	35.9	29.5	10.2	0.7
U2 - Beta(0.7, 0.3)	9.8	16.2	34.9	29.9	8.6	0.6
U2 - Beta(1, 1)	10.9	15.4	34.7	29.8	8.5	0.7
U3 - Beta(0.3, 0.7)	12	19.8	34.5	26.6	6.4	0.7
U3 - Beta(0.5, 0.5)	13.9	19.7	34.1	26	5.6	0.7
U3 - Beta(0.7, 0.3)	15.2	20.6	33.5	25	5	0.7
U3 - Beta(1, 1)	16.5	21.1	33	25.2	3.5	0.7
Mean administrations						
U2 - Beta(0.3, 0.7)	3.9	4.3	7.8	7.7	6.2	-
U2 - Beta(0.5, 0.5)	3.9	4.4	8	7.5	6.1	-
U2 - Beta(0.7, 0.3)	3.9	4.4	7.9	7.7	5.9	-
U2 - Beta(1, 1)	3.9	4.4	7.8	7.7	6.1	-
U3 - Beta(0.3, 0.7)	3.9	4.2	7.8	7.6	6.4	-
U3 - Beta(0.5, 0.5)	3.9	4.4	8	7.5	6.1	-
U3 - Beta(0.7, 0.3)	3.9	4.4	7.9	7.7	6	-
U3 - Beta(1, 1)	3.9	4.4	7.8	7.7	6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	2.5	2.9	2.1	-
U2 - Beta(0.5, 0.5)	0.4	2.9	2.5	2.8	2.1	-
U2 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.9	2	-
U2 - Beta(1, 1)	0.4	2.9	2.5	2.9	2.1	-
U3 - Beta(0.3, 0.7)	0.4	2.8	2.5	2.8	2.1	-
U3 - Beta(0.5, 0.5)	0.4	2.9	2.5	2.8	2.1	-
U3 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.9	2	-
U3 - Beta(1, 1)	0.4	2.9	2.5	2.9	2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1057
U2 - Beta(0.5, 0.5)						1055.6
U2 - Beta(0.7, 0.3)						1056.1
U2 - Beta(1, 1)						1056.3
U3 - Beta(0.3, 0.7)						1057.2
U3 - Beta(0.5, 0.5)						1056.5
U3 - Beta(0.7, 0.3)						1056.5
U3 - Beta(1, 1)						1056.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.1	7.6	16.2	42.3	25.7	0.1
U2 - Beta(0.5, 0.5)	7.9	8.8	15.7	42	25.5	0.1
U2 - Beta(0.7, 0.3)	8.9	8.7	15.6	41	25.7	0.1
U2 - Beta(1, 1)	9.2	9.4	14.9	41.8	24.6	0.1
U3 - Beta(0.3, 0.7)	11.8	9.1	15.8	40	23.2	0.1
U3 - Beta(0.5, 0.5)	12	9.7	14.7	41.6	21.9	0.1
U3 - Beta(0.7, 0.3)	13.8	10.3	15	39.2	21.6	0.1
U3 - Beta(1, 1)	13.4	10	14.2	40.6	21.7	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.4	3.6	5.4	8.1	9.5	-
U2 - Beta(0.5, 0.5)	3.5	3.6	5.3	8.1	9.4	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.4	8.2	9.1	-
U2 - Beta(1, 1)	3.5	3.7	5.3	8.2	9.3	-
U3 - Beta(0.3, 0.7)	3.4	3.5	5.4	8.1	9.5	-
U3 - Beta(0.5, 0.5)	3.5	3.6	5.4	8.1	9.3	-
U3 - Beta(0.7, 0.3)	3.5	3.7	5.4	8.2	9.2	-
U3 - Beta(1, 1)	3.5	3.7	5.3	8	9.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.7	3	2.3	-
U2 - Beta(0.5, 0.5)	0.2	2.6	1.7	3.1	2.3	-
U2 - Beta(0.7, 0.3)	0.3	2.7	1.7	3.1	2.3	-
U2 - Beta(1, 1)	0.2	2.6	1.6	3.1	2.3	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.7	3	2.4	-
U3 - Beta(0.5, 0.5)	0.2	2.6	1.7	3	2.3	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.1	2.3	-
U3 - Beta(1, 1)	0.2	2.7	1.7	3	2.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1106.7
U2 - Beta(0.5, 0.5)						1104.4
U2 - Beta(0.7, 0.3)						1103.1
U2 - Beta(1, 1)						1104.8
U3 - Beta(0.3, 0.7)						1106.3
U3 - Beta(0.5, 0.5)						1104.3
U3 - Beta(0.7, 0.3)						1103.3
U3 - Beta(1, 1)						1104
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.5	16.1	47.2	14.3	2.9	6
U2 - Beta(0.5, 0.5)	14.9	15	47.5	13.8	3	5.8
U2 - Beta(0.7, 0.3)	15.3	15.9	46.9	13.7	2.9	5.3
U2 - Beta(1, 1)	17.3	15.7	44.8	13.5	2.6	6.1
U3 - Beta(0.3, 0.7)	19	16.1	44.8	12.3	2.1	5.7
U3 - Beta(0.5, 0.5)	19.8	17.3	43.5	11.5	2.1	5.8
U3 - Beta(0.7, 0.3)	20.9	16.8	43.2	11.5	1.8	5.8
U3 - Beta(1, 1)	23.7	17	40.1	12	1.4	5.8
Mean administrations						
U2 - Beta(0.3, 0.7)	5.3	6.2	10.2	5	2.4	-
U2 - Beta(0.5, 0.5)	5.3	6.3	10.3	4.9	2.3	-
U2 - Beta(0.7, 0.3)	5.4	6.4	10.1	4.8	2.3	-
U2 - Beta(1, 1)	5.4	6.3	10	4.9	2.4	-
U3 - Beta(0.3, 0.7)	5.3	6.1	10.3	4.9	2.4	-
U3 - Beta(0.5, 0.5)	5.4	6.3	10.1	4.9	2.4	-
U3 - Beta(0.7, 0.3)	5.4	6.3	10.1	4.9	2.3	-
U3 - Beta(1, 1)	5.4	6.3	10.1	4.9	2.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.4	3.2	1.8	0.8	-
U2 - Beta(0.5, 0.5)	1	3.5	3.3	1.7	0.7	-
U2 - Beta(0.7, 0.3)	1.1	3.5	3.2	1.7	0.8	-
U2 - Beta(1, 1)	1.1	3.5	3.2	1.7	0.8	-
U3 - Beta(0.3, 0.7)	1.1	3.4	3.3	1.7	0.8	-
U3 - Beta(0.5, 0.5)	1.1	3.5	3.2	1.7	0.8	-
U3 - Beta(0.7, 0.3)	1.1	3.5	3.2	1.7	0.7	-
U3 - Beta(1, 1)	1.1	3.5	3.2	1.7	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.2
U2 - Beta(0.5, 0.5)						1032.8
U2 - Beta(0.7, 0.3)						1030.6
U2 - Beta(1, 1)						1030.1
U3 - Beta(0.3, 0.7)						1031.3
U3 - Beta(0.5, 0.5)						1030
U3 - Beta(0.7, 0.3)						1030.3
U3 - Beta(1, 1)						1030

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.1	45.9	32.7	8.9	2.2	0.2
U2 - Beta(0.5, 0.5)	11.3	45.6	31.2	9.9	1.8	0.2
U2 - Beta(0.7, 0.3)	11.6	45.3	31	10.3	1.6	0.2
U2 - Beta(1, 1)	11.4	45.1	31.7	10.3	1.3	0.2
U3 - Beta(0.3, 0.7)	13.5	49.3	28.6	7.4	1	0.2
U3 - Beta(0.5, 0.5)	14.2	49.3	27.2	8.1	1	0.2
U3 - Beta(0.7, 0.3)	15.7	49.2	25.7	8.1	1.1	0.2
U3 - Beta(1, 1)	15	50.1	26.4	7.5	0.8	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	8.9	9.5	4.8	2.3	-
U2 - Beta(0.5, 0.5)	4.5	9.1	9.2	4.9	2.3	-
U2 - Beta(0.7, 0.3)	4.6	9.3	9.1	4.8	2.2	-
U2 - Beta(1, 1)	4.5	9.1	9.2	4.9	2.2	-
U3 - Beta(0.3, 0.7)	4.5	8.9	9.3	4.9	2.3	-
U3 - Beta(0.5, 0.5)	4.5	9	9.3	4.9	2.2	-
U3 - Beta(0.7, 0.3)	4.6	9.3	9.1	4.8	2.2	-
U3 - Beta(1, 1)	4.6	9	9.2	4.9	2.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.8	2.4	1.4	0.6	-
U2 - Beta(0.5, 0.5)	0.6	3.9	2.3	1.4	0.6	-
U2 - Beta(0.7, 0.3)	0.7	3.9	2.3	1.4	0.6	-
U2 - Beta(1, 1)	0.6	3.9	2.3	1.4	0.6	-
U3 - Beta(0.3, 0.7)	0.6	3.8	2.3	1.4	0.7	-
U3 - Beta(0.5, 0.5)	0.7	3.8	2.3	1.5	0.6	-
U3 - Beta(0.7, 0.3)	0.7	3.9	2.3	1.4	0.6	-
U3 - Beta(1, 1)	0.7	3.8	2.3	1.4	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1163.3
U2 - Beta(0.5, 0.5)						1163.2
U2 - Beta(0.7, 0.3)						1162.8
U2 - Beta(1, 1)						1163.9
U3 - Beta(0.3, 0.7)						1163.7
U3 - Beta(0.5, 0.5)						1163.2
U3 - Beta(0.7, 0.3)						1162.6
U3 - Beta(1, 1)						1163.9

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.6	38	11	2.7	0.9	29.8
U2 - Beta(0.5, 0.5)	18.6	37.3	10.2	3	0.8	30.1
U2 - Beta(0.7, 0.3)	18.9	37.5	10.6	2.4	0.8	29.8
U2 - Beta(1, 1)	20.8	36.9	9.6	2.6	0.7	29.4
U3 - Beta(0.3, 0.7)	20.4	36.2	9.7	2.8	0.7	30.2
U3 - Beta(0.5, 0.5)	21.9	36.4	8.7	2.2	0.8	30
U3 - Beta(0.7, 0.3)	23.3	35.9	7.9	2.3	0.7	29.9
U3 - Beta(1, 1)	24.8	34.5	7.2	2.9	0.6	30
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	8.4	5	1.9	0.7	-
U2 - Beta(0.5, 0.5)	9	8.4	5	1.9	0.7	-
U2 - Beta(0.7, 0.3)	9	8.4	4.9	1.9	0.7	-
U2 - Beta(1, 1)	9.1	8.5	4.8	1.9	0.7	-
U3 - Beta(0.3, 0.7)	8.9	8.4	4.9	2	0.7	-
U3 - Beta(0.5, 0.5)	8.9	8.4	5	1.9	0.7	-
U3 - Beta(0.7, 0.3)	9	8.3	5	1.9	0.7	-
U3 - Beta(1, 1)	9.1	8.4	4.8	1.9	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	4	1.6	0.8	0.2	-
U2 - Beta(0.5, 0.5)	2.4	4	1.6	0.8	0.2	-
U2 - Beta(0.7, 0.3)	2.5	4	1.6	0.8	0.2	-
U2 - Beta(1, 1)	2.5	4	1.5	0.8	0.2	-
U3 - Beta(0.3, 0.7)	2.4	4	1.6	0.8	0.2	-
U3 - Beta(0.5, 0.5)	2.4	4	1.6	0.8	0.2	-
U3 - Beta(0.7, 0.3)	2.5	4	1.6	0.7	0.2	-
U3 - Beta(1, 1)	2.5	4	1.6	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						874.8
U2 - Beta(0.5, 0.5)						874.2
U2 - Beta(0.7, 0.3)						874.8
U2 - Beta(1, 1)						875.8
U3 - Beta(0.3, 0.7)						873.9
U3 - Beta(0.5, 0.5)						875.5
U3 - Beta(0.7, 0.3)						873.6
U3 - Beta(1, 1)						874.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.4	46.3	9.8	1	0.1	20.4
U2 - Beta(0.5, 0.5)	23.4	45.4	10.4	0.9	0.1	19.8
U2 - Beta(0.7, 0.3)	23.5	45.7	9.9	1	0.2	19.7
U2 - Beta(1, 1)	25.7	44	8.8	1	0.2	20.3
U3 - Beta(0.3, 0.7)	25.5	45.9	7.8	1.2	0.1	19.5
U3 - Beta(0.5, 0.5)	27.7	43.5	7.7	0.9	0	20.2
U3 - Beta(0.7, 0.3)	28.8	43.5	7.1	0.9	0	19.7
U3 - Beta(1, 1)	29.2	43.5	7.1	0.4	0.1	19.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	10.7	5.6	1.5	0.5	-
U2 - Beta(0.5, 0.5)	8.3	10.8	5.5	1.5	0.5	-
U2 - Beta(0.7, 0.3)	8.4	10.8	5.4	1.4	0.5	-
U2 - Beta(1, 1)	8.4	10.7	5.4	1.4	0.4	-
U3 - Beta(0.3, 0.7)	8.3	10.7	5.6	1.5	0.5	-
U3 - Beta(0.5, 0.5)	8.3	10.8	5.5	1.5	0.5	-
U3 - Beta(0.7, 0.3)	8.3	10.9	5.4	1.5	0.5	-
U3 - Beta(1, 1)	8.4	10.9	5.3	1.4	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	5	1.7	0.7	0.2	-
U2 - Beta(0.5, 0.5)	2.1	5	1.7	0.6	0.2	-
U2 - Beta(0.7, 0.3)	2.1	5	1.7	0.6	0.2	-
U2 - Beta(1, 1)	2.2	5	1.7	0.6	0.2	-
U3 - Beta(0.3, 0.7)	2.1	5	1.8	0.6	0.2	-
U3 - Beta(0.5, 0.5)	2.1	5	1.7	0.6	0.2	-
U3 - Beta(0.7, 0.3)	2.1	5.1	1.7	0.7	0.2	-
U3 - Beta(1, 1)	2.2	5.1	1.7	0.6	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						926.2
U2 - Beta(0.5, 0.5)						926.5
U2 - Beta(0.7, 0.3)						924.9
U2 - Beta(1, 1)						924.1
U3 - Beta(0.3, 0.7)						927.4
U3 - Beta(0.5, 0.5)						926.9
U3 - Beta(0.7, 0.3)						926.8
U3 - Beta(1, 1)						925.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12	4.1	1.2	0	0	82.7
U2 - Beta(0.5, 0.5)	14.6	4	0.9	0	0	80.5
U2 - Beta(0.7, 0.3)	14.5	4.5	0.6	0	0	80.4
U2 - Beta(1, 1)	14.8	4.5	0.4	0	0	80.3
U3 - Beta(0.3, 0.7)	11.7	4.3	0.6	0	0	83.4
U3 - Beta(0.5, 0.5)	14.6	3.8	0.6	0	0	81
U3 - Beta(0.7, 0.3)	13.8	4.9	0.6	0	0	80.7
U3 - Beta(1, 1)	14.7	4	0.7	0	0	80.6
Mean administrations						
U2 - Beta(0.3, 0.7)	11.6	4.7	1.3	0.5	0.1	-
U2 - Beta(0.5, 0.5)	11.4	4.8	1.4	0.6	0.1	-
U2 - Beta(0.7, 0.3)	11.5	4.9	1.4	0.6	0.1	-
U2 - Beta(1, 1)	11.4	4.9	1.4	0.6	0.1	-
U3 - Beta(0.3, 0.7)	11.6	4.7	1.2	0.4	0	-
U3 - Beta(0.5, 0.5)	11.6	4.8	1.4	0.6	0.1	-
U3 - Beta(0.7, 0.3)	11.4	4.8	1.4	0.6	0.1	-
U3 - Beta(1, 1)	11.5	4.8	1.4	0.6	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	2.5	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	3.1	2.6	0.5	0.3	0	-
U2 - Beta(0.7, 0.3)	3.1	2.6	0.5	0.3	0	-
U2 - Beta(1, 1)	3.1	2.6	0.5	0.3	0	-
U3 - Beta(0.3, 0.7)	3.2	2.5	0.5	0.2	0	-
U3 - Beta(0.5, 0.5)	3.2	2.6	0.5	0.3	0	-
U3 - Beta(0.7, 0.3)	3.1	2.6	0.5	0.3	0	-
U3 - Beta(1, 1)	3.2	2.6	0.4	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						635.7
U2 - Beta(0.5, 0.5)						646.3
U2 - Beta(0.7, 0.3)						649.9
U2 - Beta(1, 1)						648
U3 - Beta(0.3, 0.7)						629.7
U3 - Beta(0.5, 0.5)						647.5
U3 - Beta(0.7, 0.3)						644
U3 - Beta(1, 1)						645.6
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	32.4	36.3	26.2	1.7	0	3.4
U2 - Beta(0.5, 0.5)	27.5	24.9	37.1	7.4	0.4	2.7
U2 - Beta(0.7, 0.3)	29.9	25.7	31.6	9.4	0.7	2.7
U2 - Beta(1, 1)	27.3	24.1	38.8	6.6	0.5	2.7
U3 - Beta(0.3, 0.7)	40.8	37.2	17.8	1	0	3.2
U3 - Beta(0.5, 0.5)	31.8	26.1	33.9	5.2	0.3	2.7
U3 - Beta(0.7, 0.3)	32	24.6	32.1	8.1	0.8	2.4
U3 - Beta(1, 1)	32.3	24.8	34.9	5.3	0.2	2.5
Mean administrations						
U2 - Beta(0.3, 0.7)	13.6	10.5	5	0.4	0	-
U2 - Beta(0.5, 0.5)	10.5	8.3	8.3	2.3	0.2	-
U2 - Beta(0.7, 0.3)	10.5	7.7	7.2	3.5	0.7	-
U2 - Beta(1, 1)	10.2	8.5	8.4	2.2	0.2	-
U3 - Beta(0.3, 0.7)	14.9	10.6	3.7	0.3	0	-
U3 - Beta(0.5, 0.5)	11	8.6	7.9	1.9	0.2	-
U3 - Beta(0.7, 0.3)	10.6	8.1	7.4	3.1	0.4	-
U3 - Beta(1, 1)	11	8.6	7.9	1.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.1	4.4	1.6	0.1	0	-
U2 - Beta(0.5, 0.5)	2.7	3.7	2.4	0.7	0.1	-
U2 - Beta(0.7, 0.3)	2.7	3.5	2	1	0.2	-
U2 - Beta(1, 1)	2.6	3.8	2.4	0.7	0.1	-
U3 - Beta(0.3, 0.7)	4.6	4.5	1.2	0.1	0	-
U3 - Beta(0.5, 0.5)	2.9	3.8	2.3	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.7	3.6	2.1	0.9	0.1	-
U3 - Beta(1, 1)	2.9	3.9	2.2	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1061.1
U2 - Beta(0.5, 0.5)						1102.7
U2 - Beta(0.7, 0.3)						1109.9
U2 - Beta(1, 1)						1103.5
U3 - Beta(0.3, 0.7)						1052.7
U3 - Beta(0.5, 0.5)						1096.6
U3 - Beta(0.7, 0.3)						1105.8
U3 - Beta(1, 1)						1096.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.5	68.3	11.1	1.2	0	10.9
U2 - Beta(0.5, 0.5)	9.5	50.2	23.3	6.1	0.5	10.4
U2 - Beta(0.7, 0.3)	9.3	49.6	23.2	6.3	1	10.6
U2 - Beta(1, 1)	8.5	50.9	23.7	5.7	0.5	10.7
U3 - Beta(0.3, 0.7)	9.5	69.5	9.3	0.8	0	10.9
U3 - Beta(0.5, 0.5)	10	54.9	21	3.4	0.4	10.3
U3 - Beta(0.7, 0.3)	11	49.8	22.5	5.6	0.8	10.3
U3 - Beta(1, 1)	9.4	55.2	21.4	3.3	0.3	10.4
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	17	2.7	0.3	0	-
U2 - Beta(0.5, 0.5)	7.5	12.4	6.1	2.3	0.3	-
U2 - Beta(0.7, 0.3)	7.5	10.7	6.2	3.4	0.9	-
U2 - Beta(1, 1)	7.4	12.4	6.2	2.3	0.3	-
U3 - Beta(0.3, 0.7)	8.9	17.1	2.2	0.3	0	-
U3 - Beta(0.5, 0.5)	7.7	13.1	5.6	2	0.3	-
U3 - Beta(0.7, 0.3)	7.6	11.4	6	3	0.6	-
U3 - Beta(1, 1)	7.6	13.3	5.5	1.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	6.6	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	1.5	5.1	1.8	0.8	0.1	-
U2 - Beta(0.7, 0.3)	1.6	4.6	1.8	1.1	0.3	-
U2 - Beta(1, 1)	1.5	5.1	1.9	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.2	6.6	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	1.6	5.3	1.6	0.7	0.1	-
U3 - Beta(0.7, 0.3)	1.6	4.8	1.8	1	0.2	-
U3 - Beta(1, 1)	1.6	5.4	1.6	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1039.7
U2 - Beta(0.5, 0.5)						1057
U2 - Beta(0.7, 0.3)						1061.8
U2 - Beta(1, 1)						1056.3
U3 - Beta(0.3, 0.7)						1035.9
U3 - Beta(0.5, 0.5)						1053.7
U3 - Beta(0.7, 0.3)						1056.9
U3 - Beta(1, 1)						1054.3
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.5	7.4	36.2	48.2	1.7	2
U2 - Beta(0.5, 0.5)	5.8	7.8	22.9	53.4	8.3	1.8
U2 - Beta(0.7, 0.3)	6.5	8.5	22.4	51	9.7	1.9
U2 - Beta(1, 1)	7.2	6.4	22.4	54.7	7.5	1.8
U3 - Beta(0.3, 0.7)	7.2	10.4	36.2	43.4	0.8	2
U3 - Beta(0.5, 0.5)	9.7	7.9	21.2	53.6	5.7	1.9
U3 - Beta(0.7, 0.3)	8.9	9.7	21.5	48.6	9.5	1.8
U3 - Beta(1, 1)	9.2	9.1	20.3	54.1	5.4	1.9
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	6	10.9	8.1	0.3	-
U2 - Beta(0.5, 0.5)	4.3	4.9	7.7	10.6	2.3	-
U2 - Beta(0.7, 0.3)	4.3	5	7.3	9.6	3.6	-
U2 - Beta(1, 1)	4.3	5	7.7	10.6	2.2	-
U3 - Beta(0.3, 0.7)	4.8	6.4	10.9	7.5	0.2	-
U3 - Beta(0.5, 0.5)	4.5	5.2	7.7	10.5	2	-
U3 - Beta(0.7, 0.3)	4.6	5.3	7.2	9.7	2.9	-
U3 - Beta(1, 1)	4.8	5.3	7.4	10.4	1.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.4	3.9	2.7	0.1	-
U2 - Beta(0.5, 0.5)	0.6	3.1	2.3	3.4	0.7	-
U2 - Beta(0.7, 0.3)	0.5	3.2	2.1	3.1	1	-
U2 - Beta(1, 1)	0.5	3.2	2.3	3.4	0.7	-
U3 - Beta(0.3, 0.7)	0.9	3.5	3.9	2.5	0.1	-
U3 - Beta(0.5, 0.5)	0.6	3.3	2.3	3.4	0.6	-
U3 - Beta(0.7, 0.3)	0.7	3.3	2.1	3.1	0.9	-
U3 - Beta(1, 1)	0.8	3.3	2.2	3.3	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1047.5
U2 - Beta(0.5, 0.5)						1088.1
U2 - Beta(0.7, 0.3)						1094.9
U2 - Beta(1, 1)						1086.5
U3 - Beta(0.3, 0.7)						1039.7
U3 - Beta(0.5, 0.5)						1083.5
U3 - Beta(0.7, 0.3)						1086.5
U3 - Beta(1, 1)						1082.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.3	49.2	28.8	15.6	3.1	2
U2 - Beta(0.5, 0.5)	4.9	34.6	26.1	23	9.5	1.9
U2 - Beta(0.7, 0.3)	5.2	38.7	26.6	20.8	6.8	1.9
U2 - Beta(1, 1)	6.2	34.9	25.9	22.8	8.3	1.9
U3 - Beta(0.3, 0.7)	2.7	54	26.7	12.7	1.9	2
U3 - Beta(0.5, 0.5)	6.9	42.7	24.4	18.6	5.5	1.9
U3 - Beta(0.7, 0.3)	6.6	43.2	24.2	19.2	4.9	1.9
U3 - Beta(1, 1)	7.9	43.4	24.7	17.1	5	1.9
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	14.8	6.9	2.9	0.6	-
U2 - Beta(0.5, 0.5)	4	8.1	7.4	6.9	3.2	-
U2 - Beta(0.7, 0.3)	4.1	7.1	7.3	7.2	3.9	-
U2 - Beta(1, 1)	4.1	8.2	7.3	6.9	3.2	-
U3 - Beta(0.3, 0.7)	4.8	15.4	6.5	2.6	0.4	-
U3 - Beta(0.5, 0.5)	4.2	8.8	7.5	6.5	2.7	-
U3 - Beta(0.7, 0.3)	4.2	8	7.3	6.9	3.2	-
U3 - Beta(1, 1)	4.2	9.1	7.5	6.4	2.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	6.5	2.6	1.1	0.2	-
U2 - Beta(0.5, 0.5)	0.4	4.1	2.4	2.6	1.2	-
U2 - Beta(0.7, 0.3)	0.4	3.9	2.3	2.7	1.4	-
U2 - Beta(1, 1)	0.4	4.2	2.4	2.6	1.2	-
U3 - Beta(0.3, 0.7)	0.7	6.7	2.4	1	0.2	-
U3 - Beta(0.5, 0.5)	0.4	4.4	2.5	2.5	1	-
U3 - Beta(0.7, 0.3)	0.4	4.2	2.4	2.5	1.2	-
U3 - Beta(1, 1)	0.4	4.5	2.5	2.4	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1028.7
U2 - Beta(0.5, 0.5)						1043.7
U2 - Beta(0.7, 0.3)						1047.6
U2 - Beta(1, 1)						1044.4
U3 - Beta(0.3, 0.7)						1026.7
U3 - Beta(0.5, 0.5)						1041.9
U3 - Beta(0.7, 0.3)						1045.2
U3 - Beta(1, 1)						1040.5
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	68.9	8.2	1.3	0.2	0	21.4
U2 - Beta(0.5, 0.5)	63.6	13.1	2	0.5	0.1	20.7
U2 - Beta(0.7, 0.3)	64.6	12.8	1.7	0.6	0.1	20.2
U2 - Beta(1, 1)	63.6	12.4	2.5	0.5	0.1	20.9
U3 - Beta(0.3, 0.7)	69.5	7.9	0.7	0.2	0	21.7
U3 - Beta(0.5, 0.5)	66.3	11.6	1.2	0.5	0.1	20.3
U3 - Beta(0.7, 0.3)	65.5	11.7	1.5	0.3	0.1	20.9
U3 - Beta(1, 1)	66.2	12.1	1.2	0.3	0.1	20.1
Mean administrations						
U2 - Beta(0.3, 0.7)	22.5	3.4	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	18.9	5.7	1.4	0.5	0.1	-
U2 - Beta(0.7, 0.3)	18.4	5.9	1.6	0.7	0.1	-
U2 - Beta(1, 1)	18.8	5.7	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	22.7	3.3	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	19.5	5.5	1.2	0.4	0.1	-
U3 - Beta(0.7, 0.3)	18.8	5.6	1.5	0.6	0.1	-
U3 - Beta(1, 1)	19.6	5.5	1.2	0.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.3	1.5	0.2	0	0	-
U2 - Beta(0.5, 0.5)	4.9	2.4	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	4.8	2.4	0.5	0.3	0	-
U2 - Beta(1, 1)	4.9	2.4	0.5	0.2	0	-
U3 - Beta(0.3, 0.7)	6.4	1.4	0.1	0	0	-
U3 - Beta(0.5, 0.5)	5.1	2.3	0.4	0.2	0	-
U3 - Beta(0.7, 0.3)	4.9	2.3	0.5	0.2	0	-
U3 - Beta(1, 1)	5.2	2.3	0.4	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1016.1
U2 - Beta(0.5, 0.5)						1031.9
U2 - Beta(0.7, 0.3)						1039.6
U2 - Beta(1, 1)						1032.1
U3 - Beta(0.3, 0.7)						1013.3
U3 - Beta(0.5, 0.5)						1033.9
U3 - Beta(0.7, 0.3)						1035.5
U3 - Beta(1, 1)						1037

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21	70	4.6	0.4	0.1	3.9
U2 - Beta(0.5, 0.5)	14.9	57.1	19.8	4.5	0.5	3.2
U2 - Beta(0.7, 0.3)	15	47.3	22.8	10.2	1.5	3.2
U2 - Beta(1, 1)	14.5	57.4	20.2	4.3	0.3	3.3
U3 - Beta(0.3, 0.7)	25	68	2.6	0.3	0	4.1
U3 - Beta(0.5, 0.5)	16.6	60.9	15.4	3.4	0.3	3.4
U3 - Beta(0.7, 0.3)	16.2	51	22.1	6.7	0.8	3.2
U3 - Beta(1, 1)	15.6	63.5	14.2	3	0.3	3.4
Mean administrations						
U2 - Beta(0.3, 0.7)	12.8	15.6	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	8.8	14.6	4.8	1.2	0.1	-
U2 - Beta(0.7, 0.3)	8.5	10.8	5.9	3.4	0.9	-
U2 - Beta(1, 1)	8.7	14.7	4.8	1.2	0.1	-
U3 - Beta(0.3, 0.7)	13.3	15.4	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	9.2	15.3	4	0.9	0.1	-
U3 - Beta(0.7, 0.3)	8.7	12.5	5.6	2.4	0.3	-
U3 - Beta(1, 1)	9.2	15.5	3.8	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.7	5.4	0.3	0	0	-
U2 - Beta(0.5, 0.5)	1.9	5.3	1.3	0.4	0	-
U2 - Beta(0.7, 0.3)	1.8	4.1	1.5	1	0.3	-
U2 - Beta(1, 1)	1.8	5.3	1.4	0.4	0	-
U3 - Beta(0.3, 0.7)	3.9	5.4	0.2	0	0	-
U3 - Beta(0.5, 0.5)	2	5.5	1.1	0.3	0	-
U3 - Beta(0.7, 0.3)	1.9	4.6	1.6	0.8	0.1	-
U3 - Beta(1, 1)	2	5.6	1.1	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1104
U2 - Beta(0.5, 0.5)						1137.2
U2 - Beta(0.7, 0.3)						1143.8
U2 - Beta(1, 1)						1137.5
U3 - Beta(0.3, 0.7)						1099.6
U3 - Beta(0.5, 0.5)						1134
U3 - Beta(0.7, 0.3)						1139.7
U3 - Beta(1, 1)						1132.3
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.1	5	1.2	0.1	0	66.6
U2 - Beta(0.5, 0.5)	29.6	5.5	0.4	0.2	0	64.3
U2 - Beta(0.7, 0.3)	30.3	5	0.9	0.1	0	63.7
U2 - Beta(1, 1)	29.8	5.6	0.4	0.1	0	64.1
U3 - Beta(0.3, 0.7)	28.2	4.5	1	0	0	66.3
U3 - Beta(0.5, 0.5)	30.2	5	0.5	0.1	0	64.2
U3 - Beta(0.7, 0.3)	29.9	5.2	0.3	0.1	0	64.5
U3 - Beta(1, 1)	29.9	5.4	0.4	0.1	0	64.2
Mean administrations						
U2 - Beta(0.3, 0.7)	14.6	4.4	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	14.1	4.9	1.1	0.4	0.1	-
U2 - Beta(0.7, 0.3)	14.1	4.9	1.2	0.5	0.1	-
U2 - Beta(1, 1)	14.1	4.9	1.2	0.4	0.1	-
U3 - Beta(0.3, 0.7)	14.9	4.3	0.7	0.2	0	-
U3 - Beta(0.5, 0.5)	14.3	4.8	1.1	0.4	0.1	-
U3 - Beta(0.7, 0.3)	14.2	4.8	1.1	0.4	0.1	-
U3 - Beta(1, 1)	14.3	4.8	1.1	0.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.2	2.3	0.3	0.1	0	-
U2 - Beta(0.5, 0.5)	3.9	2.5	0.4	0.2	0	-
U2 - Beta(0.7, 0.3)	3.9	2.4	0.4	0.2	0	-
U2 - Beta(1, 1)	3.9	2.5	0.4	0.2	0	-
U3 - Beta(0.3, 0.7)	4.3	2.2	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	4	2.4	0.4	0.2	0	-
U3 - Beta(0.7, 0.3)	4	2.4	0.4	0.2	0	-
U3 - Beta(1, 1)	4	2.4	0.4	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						726.5
U2 - Beta(0.5, 0.5)						746.1
U2 - Beta(0.7, 0.3)						748.8
U2 - Beta(1, 1)						747.2
U3 - Beta(0.3, 0.7)						728.6
U3 - Beta(0.5, 0.5)						746.4
U3 - Beta(0.7, 0.3)						747.8
U3 - Beta(1, 1)						746.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.4	13.2	28.1	18.9	8.9	20.5
U2 - Beta(0.5, 0.5)	15	13.9	22.4	15.3	14.7	18.7
U2 - Beta(0.7, 0.3)	14.9	13.8	22	16.1	14.3	18.9
U2 - Beta(1, 1)	15.7	12.9	22.6	14.9	15	18.9
U3 - Beta(0.3, 0.7)	13.8	17	24.4	16.8	6.7	21.3
U3 - Beta(0.5, 0.5)	18.3	15.8	20.7	13.7	12.2	19.3
U3 - Beta(0.7, 0.3)	18.2	15.9	20.2	14.1	12.4	19.2
U3 - Beta(1, 1)	21.2	15.8	18.7	12.8	12.1	19.4
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	6.6	7.2	4.5	1.3	-
U2 - Beta(0.5, 0.5)	7.5	6.2	6	4.9	3.1	-
U2 - Beta(0.7, 0.3)	7.4	6.3	5.8	4.9	3.3	-
U2 - Beta(1, 1)	7.3	6.3	5.9	5	3.1	-
U3 - Beta(0.3, 0.7)	8.2	6.8	7.1	4.2	1	-
U3 - Beta(0.5, 0.5)	7.7	6.3	5.9	4.9	2.8	-
U3 - Beta(0.7, 0.3)	7.7	6.4	5.7	4.8	3.1	-
U3 - Beta(1, 1)	7.8	6.5	5.6	4.8	2.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	3.6	2.7	1.8	0.5	-
U2 - Beta(0.5, 0.5)	1.7	3.5	2	1.9	1	-
U2 - Beta(0.7, 0.3)	1.7	3.6	2	1.9	1	-
U2 - Beta(1, 1)	1.6	3.6	2	2	1	-
U3 - Beta(0.3, 0.7)	2.1	3.6	2.6	1.7	0.4	-
U3 - Beta(0.5, 0.5)	1.8	3.6	2	1.9	0.9	-
U3 - Beta(0.7, 0.3)	1.8	3.7	1.9	1.9	1	-
U3 - Beta(1, 1)	1.9	3.7	1.9	1.9	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						931.9
U2 - Beta(0.5, 0.5)						954.1
U2 - Beta(0.7, 0.3)						956.3
U2 - Beta(1, 1)						956.4
U3 - Beta(0.3, 0.7)						925.7
U3 - Beta(0.5, 0.5)						951.5
U3 - Beta(0.7, 0.3)						953
U3 - Beta(1, 1)						950.6
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.5	34.5	6.3	0.4	0	50.3
U2 - Beta(0.5, 0.5)	10.6	28.5	12.3	2.3	0.1	46.2
U2 - Beta(0.7, 0.3)	10	31.1	10.2	2.7	0.5	45.5
U2 - Beta(1, 1)	9.6	28.9	12.3	2.2	0.1	46.9
U3 - Beta(0.3, 0.7)	9.5	34.2	5.7	0.3	0	50.3
U3 - Beta(0.5, 0.5)	11	29.5	10.5	2	0.1	46.9
U3 - Beta(0.7, 0.3)	9.6	30.7	11.1	2.1	0.3	46.2
U3 - Beta(1, 1)	10.3	30.7	10.2	1.8	0.2	46.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10.1	11.1	1.6	0.2	0	-
U2 - Beta(0.5, 0.5)	9.6	9.2	3.4	1.1	0.2	-
U2 - Beta(0.7, 0.3)	9.4	8.4	3.7	1.8	0.4	-
U2 - Beta(1, 1)	9.5	9.2	3.5	1.1	0.2	-
U3 - Beta(0.3, 0.7)	10.2	11.1	1.5	0.1	0	-
U3 - Beta(0.5, 0.5)	9.7	9.3	3.3	1	0.1	-
U3 - Beta(0.7, 0.3)	9.6	8.7	3.5	1.5	0.2	-
U3 - Beta(1, 1)	9.7	9.4	3.3	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	4.7	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	2.2	4.1	1	0.4	0.1	-
U2 - Beta(0.7, 0.3)	2.2	3.8	1	0.6	0.1	-
U2 - Beta(1, 1)	2.2	4.1	1	0.4	0.1	-
U3 - Beta(0.3, 0.7)	2.5	4.7	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	2.3	4.1	1	0.4	0	-
U3 - Beta(0.7, 0.3)	2.2	3.9	1	0.5	0.1	-
U3 - Beta(1, 1)	2.3	4.1	1	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						837.7
U2 - Beta(0.5, 0.5)						865
U2 - Beta(0.7, 0.3)						875.5
U2 - Beta(1, 1)						866.1
U3 - Beta(0.3, 0.7)						838.6
U3 - Beta(0.5, 0.5)						864
U3 - Beta(0.7, 0.3)						870.8
U3 - Beta(1, 1)						864.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	63.7	18.2	2.4	0	0	15.7
U2 - Beta(0.5, 0.5)	65.5	17.4	1.3	0	0	15.8
U2 - Beta(0.7, 0.3)	64.9	18.6	1	0.1	0	15.4
U2 - Beta(1, 1)	64	18.6	1.6	0	0	15.8
U3 - Beta(0.3, 0.7)	65.3	17.4	1.5	0	0	15.8
U3 - Beta(0.5, 0.5)	66.6	16.7	1	0	0	15.7
U3 - Beta(0.7, 0.3)	65.7	17.9	0.9	0	0	15.5
U3 - Beta(1, 1)	67.4	16.6	1	0.1	0	14.9
Mean administrations						
U2 - Beta(0.3, 0.7)	19.1	6.8	1.6	0.2	0	-
U2 - Beta(0.5, 0.5)	17.7	7.2	2.2	0.7	0	-
U2 - Beta(0.7, 0.3)	17.5	7.3	2.2	0.8	0.1	-
U2 - Beta(1, 1)	17.5	7.4	2.3	0.7	0	-
U3 - Beta(0.3, 0.7)	20	6.4	1.2	0.2	0	-
U3 - Beta(0.5, 0.5)	18	7.1	2.1	0.7	0	-
U3 - Beta(0.7, 0.3)	17.7	7.2	2.1	0.8	0	-
U3 - Beta(1, 1)	18.1	7.1	2	0.7	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.9	3.1	0.6	0.1	0	-
U2 - Beta(0.5, 0.5)	5.2	3.4	0.7	0.3	0	-
U2 - Beta(0.7, 0.3)	5.1	3.4	0.6	0.3	0	-
U2 - Beta(1, 1)	5.1	3.4	0.7	0.3	0	-
U3 - Beta(0.3, 0.7)	6.2	3	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	5.3	3.4	0.6	0.3	0	-
U3 - Beta(0.7, 0.3)	5.2	3.4	0.6	0.3	0	-
U3 - Beta(1, 1)	5.4	3.3	0.6	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						992.4
U2 - Beta(0.5, 0.5)						1005.3
U2 - Beta(0.7, 0.3)						1008.3
U2 - Beta(1, 1)						1006.7
U3 - Beta(0.3, 0.7)						990.8
U3 - Beta(0.5, 0.5)						1003.5
U3 - Beta(0.7, 0.3)						1006.4
U3 - Beta(1, 1)						1006.3
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.7	13.5	14.3	56.4	4.3	2.8
U2 - Beta(0.5, 0.5)	11.5	14.1	13.4	46.4	12.2	2.4
U2 - Beta(0.7, 0.3)	13.4	15.3	14.4	42.1	12.5	2.3
U2 - Beta(1, 1)	13.1	14.1	12.9	44.8	12.8	2.3
U3 - Beta(0.3, 0.7)	13	16.4	11.3	53.2	3.3	2.8
U3 - Beta(0.5, 0.5)	16.5	15.4	11.8	43	10.9	2.4
U3 - Beta(0.7, 0.3)	17.4	17.2	10.5	41.5	11.1	2.3
U3 - Beta(1, 1)	17.2	16.8	12.1	40.9	10.5	2.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.9	6.5	6.8	10.3	1.2	-
U2 - Beta(0.5, 0.5)	4.7	5.8	6.3	9.4	3.6	-
U2 - Beta(0.7, 0.3)	4.8	5.8	6.1	8.5	4.4	-
U2 - Beta(1, 1)	4.9	5.9	6.2	9.2	3.6	-
U3 - Beta(0.3, 0.7)	5.3	7	6.8	9.7	0.9	-
U3 - Beta(0.5, 0.5)	5	6	6	9.3	3.4	-
U3 - Beta(0.7, 0.3)	5.3	6.1	5.9	8.5	3.9	-
U3 - Beta(1, 1)	5.4	6.1	5.9	9.1	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.7	2.4	3.7	0.4	-
U2 - Beta(0.5, 0.5)	0.7	3.6	1.9	3.3	1	-
U2 - Beta(0.7, 0.3)	0.8	3.6	1.8	3.1	1.1	-
U2 - Beta(1, 1)	0.8	3.6	1.8	3.3	1	-
U3 - Beta(0.3, 0.7)	1.1	3.8	2.4	3.5	0.3	-
U3 - Beta(0.5, 0.5)	0.9	3.6	1.8	3.3	1	-
U3 - Beta(0.7, 0.3)	1	3.7	1.7	3	1.1	-
U3 - Beta(1, 1)	1.1	3.7	1.7	3.2	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.1
U2 - Beta(0.5, 0.5)						1060.5
U2 - Beta(0.7, 0.3)						1063.7
U2 - Beta(1, 1)						1058.9
U3 - Beta(0.3, 0.7)						1025.1
U3 - Beta(0.5, 0.5)						1057
U3 - Beta(0.7, 0.3)						1056.5
U3 - Beta(1, 1)						1055.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.4	30	10.2	1.8	0.1	47.5
U2 - Beta(0.5, 0.5)	12	29	10	3.7	0.4	44.9
U2 - Beta(0.7, 0.3)	13.1	29.5	10	3.5	0.5	43.4
U2 - Beta(1, 1)	12.8	29.8	10	3.3	0.4	43.7
U3 - Beta(0.3, 0.7)	10.3	30.8	8.7	1.4	0.1	48.7
U3 - Beta(0.5, 0.5)	13.2	29.8	9.9	2.4	0.4	44.3
U3 - Beta(0.7, 0.3)	14.3	28.8	9.1	2.9	0.6	44.3
U3 - Beta(1, 1)	14.1	29.8	8.9	2.6	0.4	44.2
Mean administrations						
U2 - Beta(0.3, 0.7)	10	10.3	2.7	0.5	0	-
U2 - Beta(0.5, 0.5)	9.7	8.5	3.8	1.8	0.3	-
U2 - Beta(0.7, 0.3)	9.5	8.3	3.7	2	0.5	-
U2 - Beta(1, 1)	9.6	8.6	3.8	1.7	0.3	-
U3 - Beta(0.3, 0.7)	10.1	10.5	2.4	0.4	0	-
U3 - Beta(0.5, 0.5)	9.8	8.8	3.6	1.6	0.3	-
U3 - Beta(0.7, 0.3)	9.7	8.5	3.6	1.8	0.4	-
U3 - Beta(1, 1)	9.8	8.8	3.6	1.5	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	4.7	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	2.3	4	1.2	0.7	0.1	-
U2 - Beta(0.7, 0.3)	2.3	4	1.1	0.7	0.2	-
U2 - Beta(1, 1)	2.3	4.1	1.1	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.6	4.7	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	2.4	4.2	1.1	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.4	4	1.1	0.7	0.1	-
U3 - Beta(1, 1)	2.4	4.2	1.1	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						833.7
U2 - Beta(0.5, 0.5)						859.7
U2 - Beta(0.7, 0.3)						861.3
U2 - Beta(1, 1)						860.1
U3 - Beta(0.3, 0.7)						831.6
U3 - Beta(0.5, 0.5)						856.3
U3 - Beta(0.7, 0.3)						858.7
U3 - Beta(1, 1)						857.9
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.7	18.2	17.8	35.5	6.8	8
U2 - Beta(0.5, 0.5)	17.3	17.9	19.7	28.1	8.9	8.1
U2 - Beta(0.7, 0.3)	18.1	19.7	17.6	27	9.2	8.4
U2 - Beta(1, 1)	18.7	18.2	18	28.2	9.2	8.2
U3 - Beta(0.3, 0.7)	18	21.7	15.7	31.2	5	8.4
U3 - Beta(0.5, 0.5)	22.7	21.7	15.2	25.7	6.5	8.2
U3 - Beta(0.7, 0.3)	22.5	23	14.8	24.3	7	8.4
U3 - Beta(1, 1)	23	21.5	15.7	24.7	6.6	8.5
Mean administrations						
U2 - Beta(0.3, 0.7)	6.4	7.2	7	7.1	1.5	-
U2 - Beta(0.5, 0.5)	6.3	6.8	6.4	6.9	2.9	-
U2 - Beta(0.7, 0.3)	6.3	6.8	6.3	6.6	3.2	-
U2 - Beta(1, 1)	6.3	6.8	6.3	6.8	2.9	-
U3 - Beta(0.3, 0.7)	7	7.6	6.6	6.7	1.3	-
U3 - Beta(0.5, 0.5)	6.6	7	6.2	6.7	2.7	-
U3 - Beta(0.7, 0.3)	6.8	7	6	6.5	2.9	-
U3 - Beta(1, 1)	6.9	7.1	5.9	6.6	2.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.5	3.9	2.5	2.7	0.5	-
U2 - Beta(0.5, 0.5)	1.4	3.9	2.1	2.7	1	-
U2 - Beta(0.7, 0.3)	1.4	4	2	2.6	1	-
U2 - Beta(1, 1)	1.4	3.9	2	2.7	1	-
U3 - Beta(0.3, 0.7)	1.8	4	2.4	2.5	0.5	-
U3 - Beta(0.5, 0.5)	1.5	4	2	2.6	0.9	-
U3 - Beta(0.7, 0.3)	1.6	4	1.9	2.5	1	-
U3 - Beta(1, 1)	1.7	4.1	1.9	2.6	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						986
U2 - Beta(0.5, 0.5)						1001.5
U2 - Beta(0.7, 0.3)						1002.4
U2 - Beta(1, 1)						1000.9
U3 - Beta(0.3, 0.7)						981.3
U3 - Beta(0.5, 0.5)						997.9
U3 - Beta(0.7, 0.3)						997.2
U3 - Beta(1, 1)						995.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	37.6	32.4	11.6	3.6	0.1	14.7
U2 - Beta(0.5, 0.5)	40.6	29.8	11.4	4.2	0.3	13.7
U2 - Beta(0.7, 0.3)	40.5	29.2	11.9	4.5	0.2	13.7
U2 - Beta(1, 1)	40.8	29.9	11.1	4.1	0.4	13.7
U3 - Beta(0.3, 0.7)	44	31.5	7.7	1.8	0.1	14.9
U3 - Beta(0.5, 0.5)	44.8	30.7	7.4	2.8	0.3	14
U3 - Beta(0.7, 0.3)	44.9	30.1	7.8	3	0.2	14
U3 - Beta(1, 1)	42.5	31.8	9	2.4	0.2	14.1
Mean administrations						
U2 - Beta(0.3, 0.7)	10.5	9.1	5.5	2.7	0.6	-
U2 - Beta(0.5, 0.5)	10.7	8.7	5.3	3	0.8	-
U2 - Beta(0.7, 0.3)	10.6	8.7	5.3	3.1	0.9	-
U2 - Beta(1, 1)	10.5	8.8	5.4	3.1	0.8	-
U3 - Beta(0.3, 0.7)	11.6	9.2	4.8	2.3	0.5	-
U3 - Beta(0.5, 0.5)	11.1	8.6	5.1	2.9	0.8	-
U3 - Beta(0.7, 0.3)	11.1	8.7	5	2.9	0.8	-
U3 - Beta(1, 1)	11.2	8.7	4.9	3	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	4.4	2.1	1.1	0.3	-
U2 - Beta(0.5, 0.5)	3.1	4.4	1.8	1.3	0.3	-
U2 - Beta(0.7, 0.3)	3.1	4.5	1.8	1.4	0.3	-
U2 - Beta(1, 1)	3	4.5	1.9	1.3	0.3	-
U3 - Beta(0.3, 0.7)	3.7	4.4	1.8	0.9	0.2	-
U3 - Beta(0.5, 0.5)	3.3	4.4	1.7	1.2	0.3	-
U3 - Beta(0.7, 0.3)	3.3	4.5	1.7	1.3	0.3	-
U3 - Beta(1, 1)	3.4	4.4	1.6	1.3	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						952.8
U2 - Beta(0.5, 0.5)						960.3
U2 - Beta(0.7, 0.3)						961.8
U2 - Beta(1, 1)						962.1
U3 - Beta(0.3, 0.7)						947.9
U3 - Beta(0.5, 0.5)						959.7
U3 - Beta(0.7, 0.3)						959
U3 - Beta(1, 1)						958.9

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.2	25.5	21	20.3	12	0
U2 - Beta(0.5, 0.5)	35.1	30.2	15.5	12.8	6.4	0
U2 - Beta(0.7, 0.3)	35.4	32	14.3	11.8	6.5	0
U2 - Beta(1, 1)	37.4	30.7	14.8	10.6	6.5	0
U3 - Beta(0.3, 0.7)	29.2	28.3	19.9	15.7	6.9	0
U3 - Beta(0.5, 0.5)	39.6	32	13.8	10.4	4.2	0
U3 - Beta(0.7, 0.3)	37.7	34.6	13.7	9.9	4.1	0
U3 - Beta(1, 1)	42.5	33.2	12.3	7.9	4.1	0
Mean administrations						
U2 - Beta(0.3, 0.7)	9.2	8.3	5.6	4.7	2.2	-
U2 - Beta(0.5, 0.5)	4.4	5.3	6.1	8.2	6	-
U2 - Beta(0.7, 0.3)	4.2	5.2	6	8.4	6.2	-
U2 - Beta(1, 1)	4.3	5.5	6.3	8	5.9	-
U3 - Beta(0.3, 0.7)	9.9	8.9	5.5	4	1.7	-
U3 - Beta(0.5, 0.5)	4.7	5.8	6.2	7.9	5.4	-
U3 - Beta(0.7, 0.3)	4.8	5.7	6.2	7.9	5.3	-
U3 - Beta(1, 1)	4.9	6.1	6.3	7.7	5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4	2.4	1.9	0.9	-
U2 - Beta(0.5, 0.5)	0.6	3	2.3	3.5	2.3	-
U2 - Beta(0.7, 0.3)	0.5	3	2.2	3.7	2.2	-
U2 - Beta(1, 1)	0.5	3.1	2.4	3.4	2.2	-
U3 - Beta(0.3, 0.7)	3.1	4.2	2.3	1.7	0.7	-
U3 - Beta(0.5, 0.5)	0.7	3.2	2.3	3.4	2.1	-
U3 - Beta(0.7, 0.3)	0.8	3.2	2.3	3.4	2	-
U3 - Beta(1, 1)	0.8	3.3	2.4	3.3	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						989.5
U2 - Beta(0.5, 0.5)						1007.8
U2 - Beta(0.7, 0.3)						1010.8
U2 - Beta(1, 1)						1007
U3 - Beta(0.3, 0.7)						986.8
U3 - Beta(0.5, 0.5)						1005.5
U3 - Beta(0.7, 0.3)						1007.4
U3 - Beta(1, 1)						1005.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	47.2	50.3	1.7	0.2	0	0.6
U2 - Beta(0.5, 0.5)	20.4	68.5	10.2	0.5	0	0.4
U2 - Beta(0.7, 0.3)	21.8	62.4	13.9	1.2	0	0.7
U2 - Beta(1, 1)	19.3	69.5	10.1	0.6	0	0.5
U3 - Beta(0.3, 0.7)	51.8	47	0.6	0.1	0	0.5
U3 - Beta(0.5, 0.5)	23.6	68.7	7	0.3	0	0.4
U3 - Beta(0.7, 0.3)	23.9	61.9	12.9	0.6	0	0.7
U3 - Beta(1, 1)	22.8	70	6.5	0.3	0	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	18.6	10.8	0.5	0	0	-
U2 - Beta(0.5, 0.5)	10.1	16.2	3.3	0.4	0	-
U2 - Beta(0.7, 0.3)	9.6	13.6	5.1	1.5	0.2	-
U2 - Beta(1, 1)	9.9	16.2	3.4	0.3	0	-
U3 - Beta(0.3, 0.7)	19.1	10.5	0.3	0	0	-
U3 - Beta(0.5, 0.5)	10.8	16.4	2.5	0.3	0	-
U3 - Beta(0.7, 0.3)	9.9	14.7	4.4	0.9	0.1	-
U3 - Beta(1, 1)	10.8	16.5	2.4	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.7	3.8	0.1	0	0	-
U2 - Beta(0.5, 0.5)	2.3	5.4	1	0.1	0	-
U2 - Beta(0.7, 0.3)	2.1	4.6	1.3	0.5	0	-
U2 - Beta(1, 1)	2.2	5.4	1	0.1	0	-
U3 - Beta(0.3, 0.7)	6	3.6	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.5	5.5	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	2.2	4.9	1.2	0.3	0	-
U3 - Beta(1, 1)	2.5	5.5	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1117.5
U2 - Beta(0.5, 0.5)						1171
U2 - Beta(0.7, 0.3)						1186.7
U2 - Beta(1, 1)						1172
U3 - Beta(0.3, 0.7)						1114
U3 - Beta(0.5, 0.5)						1165.5
U3 - Beta(0.7, 0.3)						1176.9
U3 - Beta(1, 1)						1165.4
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.6	8.1	13.1	30.3	44.8	0.1
U2 - Beta(0.5, 0.5)	9.2	10.8	11	19	49.9	0.1
U2 - Beta(0.7, 0.3)	10	12.7	12	18.4	46.8	0.1
U2 - Beta(1, 1)	11.5	10.5	10	17.7	50.2	0.1
U3 - Beta(0.3, 0.7)	8	10.8	15.2	30.8	35.1	0.1
U3 - Beta(0.5, 0.5)	14.7	13.3	10.4	17.6	43.9	0.1
U3 - Beta(0.7, 0.3)	17.2	15	11.2	14.9	41.6	0.1
U3 - Beta(1, 1)	19.4	14.9	9.2	13.8	42.6	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	5.2	6.6	8.3	6	-
U2 - Beta(0.5, 0.5)	3.4	4.1	5.3	7.4	9.8	-
U2 - Beta(0.7, 0.3)	3.5	4.1	5.4	7.3	9.7	-
U2 - Beta(1, 1)	3.5	4.2	5.3	7.3	9.7	-
U3 - Beta(0.3, 0.7)	4	5.4	6.9	8.5	5.2	-
U3 - Beta(0.5, 0.5)	3.5	4.2	5.4	7.6	9.2	-
U3 - Beta(0.7, 0.3)	3.6	4.4	5.4	7.2	9.2	-
U3 - Beta(1, 1)	3.7	4.5	5.4	7.4	9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	3.1	2.7	3.3	2	-
U2 - Beta(0.5, 0.5)	0.2	2.8	1.8	3	2.9	-
U2 - Beta(0.7, 0.3)	0.2	2.9	1.8	3	2.8	-
U2 - Beta(1, 1)	0.2	2.9	1.8	3	2.9	-
U3 - Beta(0.3, 0.7)	0.5	3.2	2.8	3.4	1.7	-
U3 - Beta(0.5, 0.5)	0.2	2.9	1.9	3.1	2.7	-
U3 - Beta(0.7, 0.3)	0.3	3	1.9	3	2.7	-
U3 - Beta(1, 1)	0.3	3	1.8	3	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1016.6
U2 - Beta(0.5, 0.5)						1057.1
U2 - Beta(0.7, 0.3)						1058.2
U2 - Beta(1, 1)						1056.4
U3 - Beta(0.3, 0.7)						1010.3
U3 - Beta(0.5, 0.5)						1052.2
U3 - Beta(0.7, 0.3)						1053.8
U3 - Beta(1, 1)						1049.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.3	58.4	12.9	0.4	0	15
U2 - Beta(0.5, 0.5)	17	55	12.5	0.3	0	15.2
U2 - Beta(0.7, 0.3)	17.1	55.2	11.3	0.9	0	15.5
U2 - Beta(1, 1)	15.6	55.7	12.9	0.4	0	15.4
U3 - Beta(0.3, 0.7)	15.1	60	9.2	0.4	0	15.3
U3 - Beta(0.5, 0.5)	18.2	54.5	10.9	0.3	0	16.1
U3 - Beta(0.7, 0.3)	16.7	56.2	11.3	0.6	0	15.2
U3 - Beta(1, 1)	17	55.8	11.4	0.2	0	15.6
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	15.3	3.7	0.4	0	-
U2 - Beta(0.5, 0.5)	8.8	12.9	5.3	1.4	0.1	-
U2 - Beta(0.7, 0.3)	8.8	12.3	5.2	1.9	0.2	-
U2 - Beta(1, 1)	8.6	13	5.3	1.4	0.1	-
U3 - Beta(0.3, 0.7)	9.3	15.5	3.2	0.3	0	-
U3 - Beta(0.5, 0.5)	9	13	5.1	1.3	0.1	-
U3 - Beta(0.7, 0.3)	8.9	12.7	5	1.6	0.1	-
U3 - Beta(1, 1)	8.9	13.1	5	1.3	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	6.4	1.3	0.2	0	-
U2 - Beta(0.5, 0.5)	2	5.6	1.6	0.6	0	-
U2 - Beta(0.7, 0.3)	2	5.5	1.5	0.7	0.1	-
U2 - Beta(1, 1)	1.9	5.6	1.6	0.6	0	-
U3 - Beta(0.3, 0.7)	2.4	6.4	1.1	0.1	0	-
U3 - Beta(0.5, 0.5)	2.1	5.7	1.5	0.5	0	-
U3 - Beta(0.7, 0.3)	2.1	5.6	1.5	0.6	0	-
U3 - Beta(1, 1)	2.1	5.7	1.5	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1002.4
U2 - Beta(0.5, 0.5)						1017.3
U2 - Beta(0.7, 0.3)						1023.1
U2 - Beta(1, 1)						1018.5
U3 - Beta(0.3, 0.7)						996.9
U3 - Beta(0.5, 0.5)						1012.5
U3 - Beta(0.7, 0.3)						1022
U3 - Beta(1, 1)						1014.6

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.7	51.9	32.1	7	0.5	2.8
U2 - Beta(0.5, 0.5)	9.2	51.4	28.2	8.1	0.6	2.5
U2 - Beta(0.7, 0.3)	10.4	51.4	26.4	8.5	0.7	2.6
U2 - Beta(1, 1)	10.2	51.5	27.3	8	0.6	2.4
U3 - Beta(0.3, 0.7)	9	55.5	27.6	4.9	0.1	2.9
U3 - Beta(0.5, 0.5)	11	56.4	23.1	6.4	0.5	2.6
U3 - Beta(0.7, 0.3)	12.2	55	22.8	7	0.4	2.6
U3 - Beta(1, 1)	12.7	56.1	20.2	7.8	0.6	2.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.2	14	7.8	2.3	0.3	-
U2 - Beta(0.5, 0.5)	5.1	11.2	8.2	4.2	1	-
U2 - Beta(0.7, 0.3)	5.2	10.8	8.1	4.4	1.2	-
U2 - Beta(1, 1)	5.2	11.1	8.2	4.1	1	-
U3 - Beta(0.3, 0.7)	5.5	14.8	7.2	1.9	0.2	-
U3 - Beta(0.5, 0.5)	5.4	11.7	7.8	3.8	0.9	-
U3 - Beta(0.7, 0.3)	5.5	11.4	7.6	4.2	1	-
U3 - Beta(1, 1)	5.5	11.9	7.6	3.8	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	6.2	2.8	0.9	0.1	-
U2 - Beta(0.5, 0.5)	0.8	5.3	2.6	1.6	0.4	-
U2 - Beta(0.7, 0.3)	0.9	5.2	2.5	1.7	0.4	-
U2 - Beta(1, 1)	0.9	5.2	2.6	1.6	0.4	-
U3 - Beta(0.3, 0.7)	1	6.6	2.6	0.8	0.1	-
U3 - Beta(0.5, 0.5)	0.9	5.5	2.5	1.5	0.3	-
U3 - Beta(0.7, 0.3)	1	5.4	2.4	1.6	0.4	-
U3 - Beta(1, 1)	1	5.6	2.4	1.5	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1026.2
U2 - Beta(0.5, 0.5)						1043.4
U2 - Beta(0.7, 0.3)						1046.8
U2 - Beta(1, 1)						1044.1
U3 - Beta(0.3, 0.7)						1021.1
U3 - Beta(0.5, 0.5)						1039.6
U3 - Beta(0.7, 0.3)						1044
U3 - Beta(1, 1)						1039.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	49	0	0	0	0	51
U2 - Beta(0.5, 0.5)	45.4	0.1	0	0	0	54.5
U2 - Beta(0.7, 0.3)	42.8	0.6	0	0	0	56.6
U2 - Beta(1, 1)	45.5	0.1	0	0	0	54.4
U3 - Beta(0.3, 0.7)	49.3	0	0	0	0	50.7
U3 - Beta(0.5, 0.5)	45.5	0.1	0	0	0	54.4
U3 - Beta(0.7, 0.3)	44.4	0.4	0	0	0	55.2
U3 - Beta(1, 1)	45.6	0.1	0	0	0	54.3
Mean administrations						
U2 - Beta(0.3, 0.7)	21.2	0.4	0	0	0	-
U2 - Beta(0.5, 0.5)	19.2	1.9	0	0	0	-
U2 - Beta(0.7, 0.3)	17.6	3.1	0.1	0	0	-
U2 - Beta(1, 1)	19.2	1.9	0	0	0	-
U3 - Beta(0.3, 0.7)	21.3	0.4	0	0	0	-
U3 - Beta(0.5, 0.5)	19.3	1.8	0	0	0	-
U3 - Beta(0.7, 0.3)	18.6	2.5	0.1	0	0	-
U3 - Beta(1, 1)	19.2	1.8	0	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.5	0.2	0	0	0	-
U2 - Beta(0.5, 0.5)	3.7	0.8	0	0	0	-
U2 - Beta(0.7, 0.3)	3.3	1.1	0	0	0	-
U2 - Beta(1, 1)	3.7	0.8	0	0	0	-
U3 - Beta(0.3, 0.7)	4.5	0.2	0	0	0	-
U3 - Beta(0.5, 0.5)	3.7	0.8	0	0	0	-
U3 - Beta(0.7, 0.3)	3.5	1	0	0	0	-
U3 - Beta(1, 1)	3.7	0.8	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						930.8
U2 - Beta(0.5, 0.5)						914.6
U2 - Beta(0.7, 0.3)						904
U2 - Beta(1, 1)						916.1
U3 - Beta(0.3, 0.7)						931.7
U3 - Beta(0.5, 0.5)						914.7
U3 - Beta(0.7, 0.3)						913.8
U3 - Beta(1, 1)						913.4
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.3	64.6	7.3	0.3	0	11.5
U2 - Beta(0.5, 0.5)	17.5	60.4	8	0.9	0	13.2
U2 - Beta(0.7, 0.3)	18.2	59.5	8.3	1.3	0	12.7
U2 - Beta(1, 1)	18.3	60.3	7.5	0.9	0	13
U3 - Beta(0.3, 0.7)	18.6	63.6	6.2	0.3	0	11.3
U3 - Beta(0.5, 0.5)	18.4	61.3	6.4	1.2	0	12.7
U3 - Beta(0.7, 0.3)	18.8	59.5	7.4	1.6	0	12.7
U3 - Beta(1, 1)	20.2	58.3	7.6	1.1	0	12.8
Mean administrations						
U2 - Beta(0.3, 0.7)	8.8	15.9	3.7	0.6	0	-
U2 - Beta(0.5, 0.5)	8.9	13.6	4.9	1.5	0.1	-
U2 - Beta(0.7, 0.3)	9	13.2	4.9	1.7	0.1	-
U2 - Beta(1, 1)	8.9	13.6	4.9	1.4	0.1	-
U3 - Beta(0.3, 0.7)	9.4	16	3.2	0.5	0	-
U3 - Beta(0.5, 0.5)	9.2	13.7	4.6	1.4	0	-
U3 - Beta(0.7, 0.3)	9.2	13.3	4.8	1.6	0	-
U3 - Beta(1, 1)	9.2	13.6	4.7	1.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	6.7	1.3	0.2	0	-
U2 - Beta(0.5, 0.5)	2.1	6	1.5	0.6	0	-
U2 - Beta(0.7, 0.3)	2.2	5.9	1.4	0.7	0	-
U2 - Beta(1, 1)	2.1	6.1	1.5	0.6	0	-
U3 - Beta(0.3, 0.7)	2.5	6.7	1.2	0.2	0	-
U3 - Beta(0.5, 0.5)	2.3	6.1	1.4	0.5	0	-
U3 - Beta(0.7, 0.3)	2.3	6	1.4	0.6	0	-
U3 - Beta(1, 1)	2.3	6.1	1.4	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1015.8
U2 - Beta(0.5, 0.5)						1025.1
U2 - Beta(0.7, 0.3)						1028.2
U2 - Beta(1, 1)						1026.8
U3 - Beta(0.3, 0.7)						1010.6
U3 - Beta(0.5, 0.5)						1021.1
U3 - Beta(0.7, 0.3)						1023.9
U3 - Beta(1, 1)						1021.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.7	0.3	0	0	0	93
U2 - Beta(0.5, 0.5)	7	0.4	0	0	0	92.6
U2 - Beta(0.7, 0.3)	6.5	0.7	0	0	0	92.8
U2 - Beta(1, 1)	6.2	0.7	0	0	0	93.1
U3 - Beta(0.3, 0.7)	6.3	0.2	0	0	0	93.5
U3 - Beta(0.5, 0.5)	6.4	0.6	0	0	0	93
U3 - Beta(0.7, 0.3)	6.6	0.1	0	0	0	93.3
U3 - Beta(1, 1)	6.6	0.5	0	0	0	92.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.2	3.3	0.3	0	0	-
U2 - Beta(0.5, 0.5)	11	3.5	0.3	0	0	-
U2 - Beta(0.7, 0.3)	10.9	3.5	0.3	0	0	-
U2 - Beta(1, 1)	10.9	3.5	0.3	0	0	-
U3 - Beta(0.3, 0.7)	11.2	3.2	0.3	0	0	-
U3 - Beta(0.5, 0.5)	10.9	3.5	0.3	0	0	-
U3 - Beta(0.7, 0.3)	10.9	3.5	0.3	0	0	-
U3 - Beta(1, 1)	10.9	3.5	0.3	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	1.9	0.1	0	0	-
U2 - Beta(0.5, 0.5)	2.9	2	0.1	0	0	-
U2 - Beta(0.7, 0.3)	2.9	2	0.1	0	0	-
U2 - Beta(1, 1)	2.8	2	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3	1.9	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.9	2	0.1	0	0	-
U3 - Beta(0.7, 0.3)	2.8	2	0.1	0	0	-
U3 - Beta(1, 1)	2.9	2	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						534.4
U2 - Beta(0.5, 0.5)						531.3
U2 - Beta(0.7, 0.3)						529.2
U2 - Beta(1, 1)						529.9
U3 - Beta(0.3, 0.7)						530.5
U3 - Beta(0.5, 0.5)						529.8
U3 - Beta(0.7, 0.3)						527.6
U3 - Beta(1, 1)						529.6
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.1	0.9	11.7	75.8	11.5	0
U2 - Beta(0.5, 0.5)	0.6	0.9	2	60.6	35.9	0
U2 - Beta(0.7, 0.3)	1.9	2.2	4.9	50.1	40.9	0
U2 - Beta(1, 1)	0.8	1	2.2	60.3	35.7	0
U3 - Beta(0.3, 0.7)	0.4	1.4	13.7	76.8	7.7	0
U3 - Beta(0.5, 0.5)	1.5	1.4	3.2	63.9	30	0
U3 - Beta(0.7, 0.3)	2.4	3.2	4.5	51.8	38.1	0
U3 - Beta(1, 1)	1.7	1.9	2.4	64.8	29.2	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	4.3	7.4	13.2	1.5	-
U2 - Beta(0.5, 0.5)	3.1	3.2	3.8	13.5	6.4	-
U2 - Beta(0.7, 0.3)	3	3.1	3.8	11.3	8.8	-
U2 - Beta(1, 1)	3.1	3.2	3.9	13.5	6.3	-
U3 - Beta(0.3, 0.7)	3.7	4.4	7.7	13.2	1.1	-
U3 - Beta(0.5, 0.5)	3.1	3.2	3.9	14.1	5.7	-
U3 - Beta(0.7, 0.3)	3.1	3.2	3.9	11.8	8	-
U3 - Beta(1, 1)	3.1	3.2	4	14.2	5.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.7	3	4.6	0.5	-
U2 - Beta(0.5, 0.5)	0	2.4	1.2	4.7	1.9	-
U2 - Beta(0.7, 0.3)	0	2.4	1.1	4	2.5	-
U2 - Beta(1, 1)	0	2.5	1.2	4.7	1.9	-
U3 - Beta(0.3, 0.7)	0.4	2.8	3.1	4.6	0.4	-
U3 - Beta(0.5, 0.5)	0	2.5	1.2	4.9	1.7	-
U3 - Beta(0.7, 0.3)	0	2.5	1.2	4.1	2.3	-
U3 - Beta(1, 1)	0	2.5	1.2	4.9	1.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1040
U2 - Beta(0.5, 0.5)						1088.5
U2 - Beta(0.7, 0.3)						1096.3
U2 - Beta(1, 1)						1089.1
U3 - Beta(0.3, 0.7)						1037.3
U3 - Beta(0.5, 0.5)						1086.9
U3 - Beta(0.7, 0.3)						1090.3
U3 - Beta(1, 1)						1085.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.5	44.3	26.7	1.2	0	1.3
U2 - Beta(0.5, 0.5)	27.5	37.7	31.9	2.1	0	0.8
U2 - Beta(0.7, 0.3)	29.7	37.5	28	3.4	0	1.4
U2 - Beta(1, 1)	28	36.8	32.2	2.1	0	0.9
U3 - Beta(0.3, 0.7)	33.3	46.4	18.1	1	0	1.2
U3 - Beta(0.5, 0.5)	32.4	38	27	1.7	0	0.9
U3 - Beta(0.7, 0.3)	31	39.7	25.5	2.7	0	1.1
U3 - Beta(1, 1)	31.9	37.4	27.7	2.1	0	0.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	12	5.7	0.5	0	-
U2 - Beta(0.5, 0.5)	9.3	10.6	8.1	1.7	0.1	-
U2 - Beta(0.7, 0.3)	9.6	10.2	7.3	2.4	0.2	-
U2 - Beta(1, 1)	9.2	10.6	8.2	1.7	0.1	-
U3 - Beta(0.3, 0.7)	13	12.4	4.2	0.3	0	-
U3 - Beta(0.5, 0.5)	9.8	10.8	7.6	1.6	0.1	-
U3 - Beta(0.7, 0.3)	9.8	10.5	7.2	2.3	0.1	-
U3 - Beta(1, 1)	10	10.8	7.4	1.6	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	5	1.9	0.2	0	-
U2 - Beta(0.5, 0.5)	2.4	4.5	2.4	0.7	0	-
U2 - Beta(0.7, 0.3)	2.5	4.5	1.9	0.8	0.1	-
U2 - Beta(1, 1)	2.3	4.5	2.4	0.6	0	-
U3 - Beta(0.3, 0.7)	4.1	5.2	1.4	0.1	0	-
U3 - Beta(0.5, 0.5)	2.6	4.6	2.2	0.6	0	-
U3 - Beta(0.7, 0.3)	2.6	4.5	2	0.8	0.1	-
U3 - Beta(1, 1)	2.7	4.7	2.1	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1061.3
U2 - Beta(0.5, 0.5)						1100
U2 - Beta(0.7, 0.3)						1103.8
U2 - Beta(1, 1)						1100.4
U3 - Beta(0.3, 0.7)						1051.7
U3 - Beta(0.5, 0.5)						1094.2
U3 - Beta(0.7, 0.3)						1099.3
U3 - Beta(1, 1)						1092
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.6	35.6	29.5	1.2	0	14.1
U2 - Beta(0.5, 0.5)	22.9	35.9	25.7	2.5	0	13
U2 - Beta(0.7, 0.3)	23	36.8	23.9	3.3	0	13
U2 - Beta(1, 1)	22.3	34.8	26.5	3.2	0	13.2
U3 - Beta(0.3, 0.7)	24.1	37.2	23.6	1.3	0	13.8
U3 - Beta(0.5, 0.5)	23.9	37	23.4	2.2	0	13.5
U3 - Beta(0.7, 0.3)	25.6	37	21.4	2.4	0	13.6
U3 - Beta(1, 1)	25.2	36.1	22.7	2.7	0	13.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	11	6.7	0.7	0	-
U2 - Beta(0.5, 0.5)	9.5	9.6	7	2.1	0.2	-
U2 - Beta(0.7, 0.3)	9.7	9.7	6.4	2.4	0.2	-
U2 - Beta(1, 1)	9.2	9.8	7.1	2.1	0.2	-
U3 - Beta(0.3, 0.7)	10.8	11.2	5.7	0.5	0	-
U3 - Beta(0.5, 0.5)	9.8	9.7	6.8	1.9	0.1	-
U3 - Beta(0.7, 0.3)	9.9	9.6	6.4	2.4	0.1	-
U3 - Beta(1, 1)	9.8	9.9	6.6	1.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	4.9	2.3	0.3	0	-
U2 - Beta(0.5, 0.5)	2.4	4.5	2.1	0.8	0	-
U2 - Beta(0.7, 0.3)	2.5	4.6	1.8	0.9	0.1	-
U2 - Beta(1, 1)	2.3	4.6	2.1	0.8	0.1	-
U3 - Beta(0.3, 0.7)	3.1	5	2	0.2	0	-
U3 - Beta(0.5, 0.5)	2.6	4.6	2	0.8	0	-
U3 - Beta(0.7, 0.3)	2.6	4.6	1.8	0.9	0	-
U3 - Beta(1, 1)	2.5	4.7	1.9	0.8	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						989.2
U2 - Beta(0.5, 0.5)						1011.4
U2 - Beta(0.7, 0.3)						1016.6
U2 - Beta(1, 1)						1012.7
U3 - Beta(0.3, 0.7)						979.8
U3 - Beta(0.5, 0.5)						1008.1
U3 - Beta(0.7, 0.3)						1015.5
U3 - Beta(1, 1)						1007.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.5	32.5	48.2	2.3	0.1	5.4
U2 - Beta(0.5, 0.5)	14	30.7	43.1	6.4	0.1	5.7
U2 - Beta(0.7, 0.3)	15.7	30.6	41.8	6.2	0	5.7
U2 - Beta(1, 1)	15.3	28.5	44.8	6.2	0.1	5.1
U3 - Beta(0.3, 0.7)	16.6	35.3	40.4	1.9	0	5.8
U3 - Beta(0.5, 0.5)	17.2	30.9	41.5	4.9	0	5.5
U3 - Beta(0.7, 0.3)	18.8	29.9	40.5	5.3	0.1	5.4
U3 - Beta(1, 1)	17.7	30.4	40.8	5.6	0	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	11.1	9.7	0.9	0	-
U2 - Beta(0.5, 0.5)	7.1	9.2	9.8	2.9	0.4	-
U2 - Beta(0.7, 0.3)	7.3	9	8.8	3.6	0.6	-
U2 - Beta(1, 1)	7	9.1	9.8	3	0.4	-
U3 - Beta(0.3, 0.7)	8.6	11.6	8.5	0.6	0	-
U3 - Beta(0.5, 0.5)	7.4	9.3	9.7	2.6	0.3	-
U3 - Beta(0.7, 0.3)	7.4	9.2	9	3.4	0.4	-
U3 - Beta(1, 1)	7.5	9.4	9.5	2.6	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	4.9	3.2	0.4	0	-
U2 - Beta(0.5, 0.5)	1.6	4.3	2.9	1.1	0.1	-
U2 - Beta(0.7, 0.3)	1.7	4.3	2.5	1.2	0.2	-
U2 - Beta(1, 1)	1.6	4.3	2.9	1.1	0.1	-
U3 - Beta(0.3, 0.7)	2.3	5.2	2.8	0.3	0	-
U3 - Beta(0.5, 0.5)	1.7	4.4	2.8	1	0.1	-
U3 - Beta(0.7, 0.3)	1.7	4.3	2.6	1.1	0.1	-
U3 - Beta(1, 1)	1.8	4.4	2.8	1	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1034.9
U2 - Beta(0.5, 0.5)						1063.8
U2 - Beta(0.7, 0.3)						1068.3
U2 - Beta(1, 1)						1063.7
U3 - Beta(0.3, 0.7)						1025.1
U3 - Beta(0.5, 0.5)						1060.2
U3 - Beta(0.7, 0.3)						1065.6
U3 - Beta(1, 1)						1058

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.8	33.6	2.7	0	0	50.9
U2 - Beta(0.5, 0.5)	15.6	32.2	2.5	0.1	0	49.6
U2 - Beta(0.7, 0.3)	15	32.8	2.9	0.1	0	49.2
U2 - Beta(1, 1)	14.9	32.1	3	0.1	0	49.9
U3 - Beta(0.3, 0.7)	12.2	35.6	2.3	0	0	49.9
U3 - Beta(0.5, 0.5)	15.9	31.8	2.6	0	0	49.7
U3 - Beta(0.7, 0.3)	15	32	3.2	0.1	0	49.7
U3 - Beta(1, 1)	16.4	31.4	2.7	0.1	0	49.4
Mean administrations						
U2 - Beta(0.3, 0.7)	10.5	11.2	1.6	0.1	0	-
U2 - Beta(0.5, 0.5)	10.6	10	2.6	0.6	0	-
U2 - Beta(0.7, 0.3)	10.6	9.6	2.8	0.8	0	-
U2 - Beta(1, 1)	10.6	10	2.7	0.6	0	-
U3 - Beta(0.3, 0.7)	10.6	11.3	1.5	0.1	0	-
U3 - Beta(0.5, 0.5)	10.7	9.9	2.6	0.5	0	-
U3 - Beta(0.7, 0.3)	10.7	9.7	2.7	0.7	0	-
U3 - Beta(1, 1)	10.7	10.1	2.5	0.6	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	4.9	0.6	0.1	0	-
U2 - Beta(0.5, 0.5)	2.7	4.5	0.8	0.3	0	-
U2 - Beta(0.7, 0.3)	2.7	4.4	0.8	0.3	0	-
U2 - Beta(1, 1)	2.7	4.5	0.8	0.3	0	-
U3 - Beta(0.3, 0.7)	2.7	5	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	2.7	4.4	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	2.7	4.4	0.8	0.3	0	-
U3 - Beta(1, 1)	2.7	4.5	0.7	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						834.8
U2 - Beta(0.5, 0.5)						854
U2 - Beta(0.7, 0.3)						859.8
U2 - Beta(1, 1)						856.5
U3 - Beta(0.3, 0.7)						833.6
U3 - Beta(0.5, 0.5)						848.9
U3 - Beta(0.7, 0.3)						856.4
U3 - Beta(1, 1)						853.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	84	12.7	0.5	0	0	2.8
U2 - Beta(0.5, 0.5)	69.2	26.2	1.8	0	0	2.8
U2 - Beta(0.7, 0.3)	66.8	28.2	2.3	0	0	2.7
U2 - Beta(1, 1)	67.8	27.4	2.1	0	0	2.7
U3 - Beta(0.3, 0.7)	85.7	11.7	0	0	0	2.6
U3 - Beta(0.5, 0.5)	72.3	24	0.9	0	0	2.8
U3 - Beta(0.7, 0.3)	67.9	27.4	1.9	0.1	0	2.7
U3 - Beta(1, 1)	71.6	24.6	0.9	0	0	2.9
Mean administrations						
U2 - Beta(0.3, 0.7)	25.7	3.6	0.2	0	0	-
U2 - Beta(0.5, 0.5)	19.1	8.6	1.5	0.2	0	-
U2 - Beta(0.7, 0.3)	18.1	8.8	2.1	0.5	0	-
U2 - Beta(1, 1)	19	8.8	1.5	0.2	0	-
U3 - Beta(0.3, 0.7)	25.9	3.4	0.1	0	0	-
U3 - Beta(0.5, 0.5)	20	8.2	1.2	0.1	0	-
U3 - Beta(0.7, 0.3)	18.4	8.9	1.8	0.3	0	-
U3 - Beta(1, 1)	20.1	8.1	1	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	7.3	1.4	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.7	3.2	0.5	0.1	0	-
U2 - Beta(0.7, 0.3)	4.5	3.1	0.6	0.2	0	-
U2 - Beta(1, 1)	4.7	3.2	0.5	0.1	0	-
U3 - Beta(0.3, 0.7)	7.4	1.4	0	0	0	-
U3 - Beta(0.5, 0.5)	5.1	3.1	0.4	0	0	-
U3 - Beta(0.7, 0.3)	4.5	3.2	0.6	0.1	0	-
U3 - Beta(1, 1)	5.1	3	0.3	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1143.4
U2 - Beta(0.5, 0.5)						1160.6
U2 - Beta(0.7, 0.3)						1167.5
U2 - Beta(1, 1)						1160.5
U3 - Beta(0.3, 0.7)						1143.9
U3 - Beta(0.5, 0.5)						1157.8
U3 - Beta(0.7, 0.3)						1163.7
U3 - Beta(1, 1)						1155.1
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.5	14.4	22.9	50.4	3.2	0.6
U2 - Beta(0.5, 0.5)	15.9	13.6	16.1	45.3	8.6	0.5
U2 - Beta(0.7, 0.3)	18.6	15.7	15.8	41.2	8.3	0.4
U2 - Beta(1, 1)	16.5	13.5	15	45.2	9.4	0.4
U3 - Beta(0.3, 0.7)	15.2	19.5	21.2	41.6	2	0.5
U3 - Beta(0.5, 0.5)	22.2	15.3	15.2	39.9	6.9	0.5
U3 - Beta(0.7, 0.3)	21.8	18.7	13.6	37.9	7.6	0.4
U3 - Beta(1, 1)	23.3	17.5	13.2	38.4	7.2	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5.9	7.4	7.9	8	0.7	-
U2 - Beta(0.5, 0.5)	4.5	5.6	6.6	10.1	3.2	-
U2 - Beta(0.7, 0.3)	4.8	5.5	6.6	9.3	3.8	-
U2 - Beta(1, 1)	4.6	5.6	6.5	10.1	3.1	-
U3 - Beta(0.3, 0.7)	6.7	8.3	7.5	6.9	0.5	-
U3 - Beta(0.5, 0.5)	4.9	5.7	6.5	10.1	2.8	-
U3 - Beta(0.7, 0.3)	5.2	5.8	6.5	9.3	3.2	-
U3 - Beta(1, 1)	5	5.9	6.4	9.9	2.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	3.8	3.1	2.7	0.3	-
U2 - Beta(0.5, 0.5)	0.6	3.2	2.1	3.3	1	-
U2 - Beta(0.7, 0.3)	0.7	3.3	2.1	3.1	1.1	-
U2 - Beta(1, 1)	0.7	3.3	2.1	3.3	1	-
U3 - Beta(0.3, 0.7)	1.7	4.1	2.9	2.4	0.2	-
U3 - Beta(0.5, 0.5)	0.8	3.3	2.1	3.4	0.9	-
U3 - Beta(0.7, 0.3)	0.9	3.4	2	3.1	1	-
U3 - Beta(1, 1)	0.9	3.4	2	3.3	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1029.5
U2 - Beta(0.5, 0.5)						1075.7
U2 - Beta(0.7, 0.3)						1080
U2 - Beta(1, 1)						1074.8
U3 - Beta(0.3, 0.7)						1018
U3 - Beta(0.5, 0.5)						1072.3
U3 - Beta(0.7, 0.3)						1071.3
U3 - Beta(1, 1)						1069.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.9	78.8	11.1	1.2	0	0
U2 - Beta(0.5, 0.5)	2.3	45.1	31.3	19.3	2	0
U2 - Beta(0.7, 0.3)	5	35.6	26.4	26.4	6.6	0
U2 - Beta(1, 1)	2.3	43.8	32.9	19	2	0
U3 - Beta(0.3, 0.7)	12.3	79.7	7.1	0.9	0	0
U3 - Beta(0.5, 0.5)	3.8	50.8	27.9	16.6	0.9	0
U3 - Beta(0.7, 0.3)	5	39.2	28.2	24.1	3.5	0
U3 - Beta(1, 1)	3.6	51	29.4	15	1	0
Mean administrations						
U2 - Beta(0.3, 0.7)	9	18.8	2.1	0.2	0	-
U2 - Beta(0.5, 0.5)	3.9	14.3	7.6	3.7	0.5	-
U2 - Beta(0.7, 0.3)	3.6	7.2	7.8	8.2	3.2	-
U2 - Beta(1, 1)	3.9	14.2	7.7	3.7	0.6	-
U3 - Beta(0.3, 0.7)	9.4	19	1.5	0.2	0	-
U3 - Beta(0.5, 0.5)	4.1	15.3	7	3.2	0.4	-
U3 - Beta(0.7, 0.3)	4	9.6	7.9	6.6	1.9	-
U3 - Beta(1, 1)	4.1	15.3	7.2	3.1	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	6.8	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	0.3	5.4	2.3	1.1	0.2	-
U2 - Beta(0.7, 0.3)	0.2	3.2	2.2	2.3	1	-
U2 - Beta(1, 1)	0.3	5.4	2.4	1.1	0.2	-
U3 - Beta(0.3, 0.7)	2.8	6.8	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	0.4	5.8	2.2	1	0.1	-
U3 - Beta(0.7, 0.3)	0.4	4	2.3	1.9	0.6	-
U3 - Beta(1, 1)	0.4	5.8	2.2	1	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1097.1
U2 - Beta(0.5, 0.5)						1139
U2 - Beta(0.7, 0.3)						1164.8
U2 - Beta(1, 1)						1138.5
U3 - Beta(0.3, 0.7)						1095.1
U3 - Beta(0.5, 0.5)						1135.4
U3 - Beta(0.7, 0.3)						1152.6
U3 - Beta(1, 1)						1135
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3	9.7	49.5	31.9	5.9	0.1
U2 - Beta(0.5, 0.5)	11.6	13.7	33.9	31.5	9.1	0.2
U2 - Beta(0.7, 0.3)	12.2	16.9	34.6	28.8	7.4	0.1
U2 - Beta(1, 1)	12.2	15.3	33.7	30.4	8.3	0.1
U3 - Beta(0.3, 0.7)	6.1	13.4	50.7	26.7	3	0.1
U3 - Beta(0.5, 0.5)	15.5	17.9	34	27.1	5.4	0.1
U3 - Beta(0.7, 0.3)	16.2	20.8	33.5	24	5.3	0.2
U3 - Beta(1, 1)	17.3	18.8	32.7	26.6	4.5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	6.6	12.1	5.8	1.1	-
U2 - Beta(0.5, 0.5)	3.5	4.5	8.9	8.9	4.2	-
U2 - Beta(0.7, 0.3)	3.6	4.5	8.7	8.7	4.4	-
U2 - Beta(1, 1)	3.6	4.6	9	8.8	4.1	-
U3 - Beta(0.3, 0.7)	4.6	7.1	12.1	5.3	0.9	-
U3 - Beta(0.5, 0.5)	3.6	4.7	9.4	8.7	3.6	-
U3 - Beta(0.7, 0.3)	3.8	5	8.9	8.4	3.9	-
U3 - Beta(1, 1)	3.8	5.1	9.2	8.6	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.6	4.6	2.1	0.4	-
U2 - Beta(0.5, 0.5)	0.2	2.9	2.9	3.4	1.4	-
U2 - Beta(0.7, 0.3)	0.3	3	2.8	3.3	1.4	-
U2 - Beta(1, 1)	0.2	2.9	2.9	3.3	1.4	-
U3 - Beta(0.3, 0.7)	0.8	3.8	4.6	2	0.3	-
U3 - Beta(0.5, 0.5)	0.3	3	3.1	3.3	1.2	-
U3 - Beta(0.7, 0.3)	0.4	3.2	2.9	3.1	1.3	-
U3 - Beta(1, 1)	0.4	3.2	3	3.2	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1022.8
U2 - Beta(0.5, 0.5)						1054.2
U2 - Beta(0.7, 0.3)						1057.9
U2 - Beta(1, 1)						1052.3
U3 - Beta(0.3, 0.7)						1018.4
U3 - Beta(0.5, 0.5)						1050.1
U3 - Beta(0.7, 0.3)						1052.4
U3 - Beta(1, 1)						1048.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.1	4.1	20.9	62.2	9.6	0.1
U2 - Beta(0.5, 0.5)	7.8	6.1	13.6	51.8	20.6	0.1
U2 - Beta(0.7, 0.3)	8.3	7.8	12.7	48.3	22.8	0.1
U2 - Beta(1, 1)	8.2	6.1	12.6	52.3	20.7	0.1
U3 - Beta(0.3, 0.7)	6.8	5.2	21	60.8	6.1	0.1
U3 - Beta(0.5, 0.5)	12	6.6	13.3	50.3	17.7	0.1
U3 - Beta(0.7, 0.3)	12.8	9.7	12.9	46.4	18.1	0.1
U3 - Beta(1, 1)	12.9	8.6	12	50.2	16.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.6	8.7	11.3	1.6	-
U2 - Beta(0.5, 0.5)	3.3	3.8	6.3	11.2	5.4	-
U2 - Beta(0.7, 0.3)	3.4	3.8	6.2	10.3	6.4	-
U2 - Beta(1, 1)	3.4	3.8	6.2	11.3	5.2	-
U3 - Beta(0.3, 0.7)	3.8	4.8	8.9	11.2	1.3	-
U3 - Beta(0.5, 0.5)	3.5	3.9	6.4	11.5	4.8	-
U3 - Beta(0.7, 0.3)	3.6	4.1	6.2	10.4	5.7	-
U3 - Beta(1, 1)	3.6	4.2	6.2	11.3	4.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.9	3.3	3.9	0.5	-
U2 - Beta(0.5, 0.5)	0.1	2.7	1.9	4	1.5	-
U2 - Beta(0.7, 0.3)	0.2	2.8	1.8	3.7	1.6	-
U2 - Beta(1, 1)	0.2	2.8	1.9	4	1.5	-
U3 - Beta(0.3, 0.7)	0.4	3	3.4	3.9	0.4	-
U3 - Beta(0.5, 0.5)	0.2	2.8	1.9	4	1.3	-
U3 - Beta(0.7, 0.3)	0.3	2.9	1.9	3.7	1.6	-
U3 - Beta(1, 1)	0.3	2.9	1.8	4	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1047.4
U2 - Beta(0.5, 0.5)						1087.4
U2 - Beta(0.7, 0.3)						1091.4
U2 - Beta(1, 1)						1086.7
U3 - Beta(0.3, 0.7)						1042.9
U3 - Beta(0.5, 0.5)						1084.1
U3 - Beta(0.7, 0.3)						1083.7
U3 - Beta(1, 1)						1081.3
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	14.5	56.6	14.2	1.6	5.2
U2 - Beta(0.5, 0.5)	12.6	15.9	49.3	14.8	2.4	5
U2 - Beta(0.7, 0.3)	13.5	17.1	46.8	15.5	2.6	4.5
U2 - Beta(1, 1)	12.6	16.7	48.5	14.9	2.6	4.7
U3 - Beta(0.3, 0.7)	10.4	16.9	53.2	13.2	0.9	5.4
U3 - Beta(0.5, 0.5)	16.3	18.9	44.3	13.5	2	5
U3 - Beta(0.7, 0.3)	17.2	18.1	43.2	14.9	2.1	4.5
U3 - Beta(1, 1)	18.8	17.7	40.5	16.1	1.8	5.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5.4	7.6	12.3	3.8	0.4	-
U2 - Beta(0.5, 0.5)	5.4	7	10.3	5.4	1.5	-
U2 - Beta(0.7, 0.3)	5.4	7	9.8	5.6	1.9	-
U2 - Beta(1, 1)	5.4	7.1	10.2	5.4	1.4	-
U3 - Beta(0.3, 0.7)	5.8	7.8	12.2	3.4	0.3	-
U3 - Beta(0.5, 0.5)	5.7	7.2	10.2	5.3	1.3	-
U3 - Beta(0.7, 0.3)	5.7	7.2	9.6	5.5	1.6	-
U3 - Beta(1, 1)	6	7.3	9.8	5.2	1.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.9	4.2	1.4	0.1	-
U2 - Beta(0.5, 0.5)	1	3.8	3.2	2	0.5	-
U2 - Beta(0.7, 0.3)	1	3.8	3	2	0.6	-
U2 - Beta(1, 1)	1	3.9	3.2	2	0.5	-
U3 - Beta(0.3, 0.7)	1.3	3.9	4.2	1.3	0.1	-
U3 - Beta(0.5, 0.5)	1.2	3.9	3.2	1.9	0.5	-
U3 - Beta(0.7, 0.3)	1.2	3.9	3	1.9	0.6	-
U3 - Beta(1, 1)	1.3	4	3	1.9	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1029.5
U2 - Beta(0.5, 0.5)						1044.9
U2 - Beta(0.7, 0.3)						1047.7
U2 - Beta(1, 1)						1043.9
U3 - Beta(0.3, 0.7)						1025.9
U3 - Beta(0.5, 0.5)						1041.7
U3 - Beta(0.7, 0.3)						1042.9
U3 - Beta(1, 1)						1039.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.2	74.2	13.6	0.9	0	0.1
U2 - Beta(0.5, 0.5)	6.4	47.8	36.5	8.2	1	0.1
U2 - Beta(0.7, 0.3)	8.2	45.5	30.5	14.3	1.4	0.1
U2 - Beta(1, 1)	7.2	48.7	35.4	7.8	0.8	0.1
U3 - Beta(0.3, 0.7)	13.8	76	9.4	0.7	0	0.1
U3 - Beta(0.5, 0.5)	9.1	53.1	30.2	6.8	0.7	0.1
U3 - Beta(0.7, 0.3)	10.8	48	29.6	10.2	1.3	0.1
U3 - Beta(1, 1)	10.2	53.7	29.5	5.8	0.7	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	9.2	17.9	2.7	0.2	0	-
U2 - Beta(0.5, 0.5)	4.3	13.7	8.7	2.8	0.5	-
U2 - Beta(0.7, 0.3)	4.4	9.8	9	5.3	1.5	-
U2 - Beta(1, 1)	4.4	13.8	8.5	2.8	0.5	-
U3 - Beta(0.3, 0.7)	9.5	18.3	2	0.2	0	-
U3 - Beta(0.5, 0.5)	4.7	14.7	7.9	2.3	0.4	-
U3 - Beta(0.7, 0.3)	4.6	11.7	8.7	4	0.9	-
U3 - Beta(1, 1)	4.7	14.9	7.8	2.2	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	6.6	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	0.5	5.3	2.5	0.9	0.1	-
U2 - Beta(0.7, 0.3)	0.5	4	2.4	1.6	0.4	-
U2 - Beta(1, 1)	0.5	5.3	2.4	0.9	0.2	-
U3 - Beta(0.3, 0.7)	2.9	6.8	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	0.6	5.7	2.2	0.8	0.1	-
U3 - Beta(0.7, 0.3)	0.6	4.6	2.4	1.2	0.3	-
U3 - Beta(1, 1)	0.6	5.8	2.2	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1086.7
U2 - Beta(0.5, 0.5)						1143.8
U2 - Beta(0.7, 0.3)						1161.4
U2 - Beta(1, 1)						1143.9
U3 - Beta(0.3, 0.7)						1083.9
U3 - Beta(0.5, 0.5)						1136.5
U3 - Beta(0.7, 0.3)						1149.6
U3 - Beta(1, 1)						1136
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.3	39.3	13.8	2.7	0.3	29.6
U2 - Beta(0.5, 0.5)	15.1	39.7	12.8	3.5	0.4	28.5
U2 - Beta(0.7, 0.3)	16.5	39.5	11.9	3.3	0.6	28.2
U2 - Beta(1, 1)	16.2	39.6	12.2	3.4	0.5	28.1
U3 - Beta(0.3, 0.7)	16.7	40.2	10	2.3	0.2	30.6
U3 - Beta(0.5, 0.5)	17.6	40.2	10.3	2.7	0.3	28.9
U3 - Beta(0.7, 0.3)	17.1	39.9	11.1	2.9	0.4	28.6
U3 - Beta(1, 1)	19.6	38.2	10	3.4	0.4	28.4
Mean administrations						
U2 - Beta(0.3, 0.7)	9	11.5	4.5	1.4	0.2	-
U2 - Beta(0.5, 0.5)	8.8	10.2	4.8	2.4	0.5	-
U2 - Beta(0.7, 0.3)	8.7	10.1	4.7	2.7	0.6	-
U2 - Beta(1, 1)	8.8	10.2	4.8	2.4	0.5	-
U3 - Beta(0.3, 0.7)	9.3	11.8	4	1.2	0.1	-
U3 - Beta(0.5, 0.5)	9.1	10.3	4.6	2.3	0.4	-
U3 - Beta(0.7, 0.3)	9.1	10.2	4.5	2.4	0.5	-
U3 - Beta(1, 1)	9.2	10.4	4.4	2.3	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	5.4	1.6	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.1	5	1.5	1	0.2	-
U2 - Beta(0.7, 0.3)	2.1	5	1.5	1	0.2	-
U2 - Beta(1, 1)	2.1	5	1.5	1	0.2	-
U3 - Beta(0.3, 0.7)	2.4	5.5	1.4	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.2	5	1.5	0.9	0.2	-
U3 - Beta(0.7, 0.3)	2.2	5	1.4	0.9	0.2	-
U3 - Beta(1, 1)	2.2	5.1	1.4	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						919.4
U2 - Beta(0.5, 0.5)						935.3
U2 - Beta(0.7, 0.3)						938.1
U2 - Beta(1, 1)						936.1
U3 - Beta(0.3, 0.7)						910.8
U3 - Beta(0.5, 0.5)						930.9
U3 - Beta(0.7, 0.3)						932.5
U3 - Beta(1, 1)						933.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.8	45.9	13.6	1.2	0.2	24.3
U2 - Beta(0.5, 0.5)	15.1	48.9	11.2	1	0	23.8
U2 - Beta(0.7, 0.3)	15.2	48.2	12.2	1.7	0	22.7
U2 - Beta(1, 1)	14.9	47.7	12.1	1.4	0	23.9
U3 - Beta(0.3, 0.7)	18	46.5	9.7	0.9	0	24.9
U3 - Beta(0.5, 0.5)	17.5	47.9	10.1	0.9	0	23.6
U3 - Beta(0.7, 0.3)	17	47.4	11.3	1.4	0	22.9
U3 - Beta(1, 1)	19.4	45.5	9.9	1.6	0	23.6
Mean administrations						
U2 - Beta(0.3, 0.7)	8.7	12.9	4.6	1.2	0.1	-
U2 - Beta(0.5, 0.5)	8.6	11.7	4.9	2.1	0.4	-
U2 - Beta(0.7, 0.3)	8.6	11.5	5	2.3	0.5	-
U2 - Beta(1, 1)	8.5	11.7	4.9	2.1	0.4	-
U3 - Beta(0.3, 0.7)	9	13.3	4.1	1.1	0.1	-
U3 - Beta(0.5, 0.5)	8.9	11.8	4.7	2	0.4	-
U3 - Beta(0.7, 0.3)	8.8	11.6	4.8	2.1	0.4	-
U3 - Beta(1, 1)	8.9	11.8	4.6	2	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	5.9	1.7	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2	5.6	1.5	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2	5.6	1.5	0.9	0.2	-
U2 - Beta(1, 1)	2	5.7	1.6	0.9	0.1	-
U3 - Beta(0.3, 0.7)	2.3	6	1.5	0.4	0.1	-
U3 - Beta(0.5, 0.5)	2.2	5.7	1.5	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.1	5.6	1.5	0.9	0.1	-
U3 - Beta(1, 1)	2.2	5.7	1.4	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						947.1
U2 - Beta(0.5, 0.5)						960
U2 - Beta(0.7, 0.3)						963.7
U2 - Beta(1, 1)						960.4
U3 - Beta(0.3, 0.7)						943.9
U3 - Beta(0.5, 0.5)						959.7
U3 - Beta(0.7, 0.3)						960.2
U3 - Beta(1, 1)						958.6